Lattice Yang–Mills theory at strong coupling II: the infinite-volume theory, its transfer operator and its mass gap

A sequel to the foundations monograph of the project "Solving Yang–Mills and the Mass Gap"

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Abstract

We construct the infinite-volume lattice Yang–Mills theory on $\mathbb{Z}^d$ at strong coupling and prove that it has a unique vacuum and a mass gap, with complete proofs and explicit constants, in the framework of the foundations monograph [F] of this project. For every $\beta \geq 0$, every infinite-volume limit point of the periodic Wilson measures is a translation-invariant, reflection-positive Gibbs measure; its Osterwalder–Schrader reconstruction gives a Hilbert space and a self-adjoint contraction $\hat{T}$ (the transfer operator), so that the mass gap $m(\mu)$ of an infinite-volume lattice theory is defined at every coupling. For $0 < \beta < \beta_*(d) := \operatorname{artanh} \frac{1}{6(d-1)}$ ($\beta_*(4) \approx 0.0556$) we prove: the Gibbs measure μ_β is unique; periodic, free and fixed-boundary finite-volume measures all converge to it at an explicit exponential rate; μ_β is invariant under translations, the hypercubic group and *all* gauge transformations; it clusters exponentially; and it has the global Markov property with respect to time slices. Consequently the spatial link configurations of successive time slices form a stationary reversible Markov chain whose transition operator P on $L^2(\nu)$, ν the time-zero marginal (the infinite-volume vacuum measure), is unitarily equivalent to $\hat{T}$, satisfies $0 \leq P \leq 1$, and has spectrum in $[0, \alpha_*(\beta)] \cup \{1\}$ with 1 a simple eigenvalue, where $\alpha_*(\beta) = 6(d-1)\tanh\beta$. Hence $m(\mu_\beta) = -\log(1 - \operatorname{gap} P) \geq \log(1/\alpha_*(\beta))$: the exact reformulation of the gap as a Markov-chain spectral gap holds in infinite volume. The threshold improves the Dobrushin threshold of [F] by a factor of about four through a sharp total-variation estimate, which also improves the volume-uniform finite-volume gap of [F]. This completes milestone T1.1 of the research program. Nothing here uses the non-abelian structure of G: every statement holds verbatim for $G = U(1)$, so these results describe the strong-coupling endpoint only.

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1 Introduction and main results

The foundations monograph [F] develops lattice Yang–Mills theory on a finite periodic lattice: Wilson's measure, reflection positivity, the transfer matrix and its unique vacuum, the reduction of the continuum problem to lattice hypotheses, a mass gap uniform in the volume for $\beta < \beta_D(d)$, and the polymer representation; the sequel [III] proves the area law at strong coupling, for every infinite-volume limit point. A quantum field theory, however, lives in infinite volume, and the research program (`docs/03-research-program.md`, milestone T1.1) asks for the infinite-volume lattice theory at strong coupling: existence and uniqueness of the infinite-volume limit, translation invariance, and the Osterwalder–Schrader (OS) axioms of the lattice theory. This article supplies these, and goes further in one respect that matters for the program: it identifies the transfer operator of the infinite-volume theory with the transition operator of an explicit reversible Markov chain (the ground-state chain of [F, Theorem 'Exact reformulation of the gap' (`thm:chain`)], now in infinite volume) and proves that this chain has a spectral gap.

Convention 1.1 (Labels). As in [F, Convention 'Labels']: every Theorem, Proposition, Lemma and Corollary below has a complete proof here or in [F]; Cited Theorems are standard facts from the literature with a reference; Remarks are commentary. Constants are explicit and not optimised.

Throughout, G, ρ, N, Tr, dU and the plaquette weight $w_\beta(U) = \exp(\frac{\beta}{N}\mathrm{Re}\,\mathrm{Tr}\,U)$ are as in [F] ([F, Convention 'Group'], [F, eq. `eq:w`]), with the additional assumption that G is metrisable (true for every compact Lie group, e.g. $SU(N)$, $U(N)$, $U(1)$). We use the constants

$$\alpha_*(\beta) := 6(d-1)\tanh\beta, \qquad \beta_*(d) := \operatorname{artanh}\frac{1}{6(d-1)}, \qquad \text{so that } \alpha_*(\beta) < 1 \iff \beta < \beta_*(d). \tag{1}$$

Numerically $\beta_*(2) \approx 0.1682$, $\beta_*(3) \approx 0.0835$, $\beta_*(4) \approx 0.0556$, against $\beta_D(d) = \frac{1}{4}\log(1 + \frac{1}{6(d-1)})$ of [F, Theorem 'Strong-coupling clustering and mass gap, uniform in the volume' (`thm:strong`)]: $\beta_D(4) \approx 0.0135$. The objects in the next two theorems are defined in Sections 2–7.

Theorem 1.2 (Infinite-volume limit points, every β)**.** *Let $d \geq 2$, $\beta \geq 0$, and let Λ_k be periodic lattices whose sides all tend to infinity and whose time extents $L_0^{(k)}$ are even. Then the periodised Wilson measures $\tilde{\mu}_{\Lambda_k,\beta}$ on $G^{E(\mathbb{Z}^d)}$ have weak limit points, and every limit point μ is a Gibbs measure for the Wilson specification, invariant under all translations of $\mathbb{Z}^d$ and under the time reflection, and reflection positive. Its OS reconstruction yields a Hilbert space $\mathcal{H}_\mu$, a unit vector Ω_μ and a self-adjoint operator $\hat{T}_\mu$ with $\|\hat{T}_\mu\| \leq 1$ and $\hat{T}_\mu\Omega_\mu = \Omega_\mu$; the* mass gap $m(\mu) := -\log\|\hat{T}_\mu(1 - |\Omega_\mu\rangle\langle\Omega_\mu|)\| \in [0,\infty]$ *is therefore defined.*

Theorem 1.3 (The infinite-volume theory at strong coupling)**.** *Let $d \geq 2$, $0 < \beta < \beta_*(d)$ and $\alpha := \alpha_*(\beta) < 1$.*

(i) (Uniqueness, convergence) *There is exactly one Gibbs measure μ_β. If F is bounded, measurable and depends only on links with both endpoints in the box $\{|x|_\infty \le r\}$, and $n \ge r$, then*

$$|\langle F\rangle - \mu_\beta(F)| \le \frac{\|\delta(F)\|_1}{1-\alpha}\,\alpha^{n+1-r}$$

for the periodised Wilson measure of every periodic lattice with all sides $\ge 2n+3$, for the free-boundary measure in $\{|x|_\infty \le n'\}$ with $n' \ge n+1$, and for the measure with arbitrary fixed boundary condition outside the links of $\{|x|_\infty \le n'\}$, $n' \ge n$. In particular every sequence of periodic measures with all sides tending to infinity converges to μ_β on every bounded measurable local function.

(ii) (Symmetry) *μ_β is invariant under translations, the hypercubic group, and every gauge transformation $g \in G^{\mathbb{Z}^d}$; the link variables of any finite forest are independent and Haar distributed.*

(iii) (Clustering) *For bounded measurable f, g depending on disjoint finite link sets A, B, $|\mathrm{Cov}_{\mu_\beta}(f,g)| \le \|g\|_\infty \|\delta(f)\|_1\, \alpha^{\varrho(A,B)}/(1-\alpha)$.*

(iv) (Markov property, ground-state chain) *The time slices $U^{(t)}$ ($t \in \mathbb{Z}$) of spatial links form a stationary reversible Markov chain with stationary law ν (the time-zero marginal of μ_β). Its transition operator P on $L^2(\nu)$ is unitarily equivalent to the transfer operator $\hat{T}_{\mu_\beta}$ and satisfies $0 \le P \le 1$.*

(v) (Mass gap) $\operatorname{spec} P \subseteq [0,\alpha] \cup \{1\}$, *the eigenvalue 1 is simple (unique vacuum), and*

$$m(\mu_\beta) = -\log(1 - \operatorname{gap} P) \;\ge\; \log\frac{1}{\alpha_*(\beta)} \;\longrightarrow\; \infty \quad (\beta \downarrow 0).$$

For all $A, B \in L^2(\nu)$ and $t \in \mathbb{N}$, $\big|\mu_\beta(\overline{B(U^{(0)})}A(U^{(t)})) - \overline{\nu(B)}\nu(A)\big| \le \alpha_*(\beta)^t \|B - \nu(B)\|_{L^2(\nu)}\|A - \nu(A)\|_{L^2(\nu)}$.

(vi) (Finite volume) *For every periodic lattice with all sides ≥ 3, the finite-volume gap of [F] satisfies $m_{\Lambda_s}(\beta) \ge \log(1/\alpha_*(\beta))$.*

Theorem 1.2 collects Proposition 4.3, Proposition 6.4(a) and Theorem 6.5. Theorem 1.3 collects Theorem 4.4, Theorem 4.6, Proposition 4.7, Theorem 5.1, Theorem 5.3, Corollary 5.4, Theorem 7.1 and Corollary 3.4. Statement (v) is the infinite-volume form of the exact reformulation [F, Theorem 'Exact reformulation of the gap' (`thm:chain`)] (`docs/03-research-program.md`, §3.5): the mass gap of the infinite-volume lattice theory equals $-\log(1-\operatorname{gap} P)$ for the reversible Markov chain of time slices of the vacuum.

Strategy. The single-link conditional distributions of the Wilson specification depend on a neighbouring link only through one plaquette, so two such distributions have a density ratio in an interval $[ce^{-2\beta}, ce^{2\beta}]$; a sharp elementary inequality (Lemma 3.1) turns this into a total-variation bound $\tanh\beta$, hence a Dobrushin constant $\alpha_*(\beta)$. The coupling argument of [F, Theorem 'Covariance bound under Dobrushin's condition' (`thm:dobrushin`)], in a form that tracks which boundary links differ (Lemma 3.5), shows that conditional expectations in a box forget the boundary condition exponentially fast. Since every finite-volume measure (periodic, free, fixed) satisfies the DLR equation in boxes well inside its volume, this gives uniqueness, convergence with a rate, clustering, and—applied to half-boxes—the global Markov property with respect to a time slice. Reflection positivity passes to the limit; the Markov property identifies the OS Hilbert space with $L^2(\nu)$ and the OS transfer operator with the transition operator P of the slice chain; positivity of P comes from positivity of the finite-volume transfer matrices of [F]; and the spectral gap of P follows from clustering.

What is used from [F] and [III]. From [F]: the conventions, [F, eq. `eq:w`] and [F, eq. `eq:plaquette`]; Theorem 'Basic properties' (`thm:basic`); Lemma 'Reflection of configurations and functions' and Theorem 'Reflection positivity' (`lem:reflsym`, `thm:rp`); Lemma `lem:K` and Theorem 'Transfer matrix formalism' (`thm:tm`); Definition 'Finite spin system' (`def:spin`), the interpolation inequalities (`eq:interp`), Lemma 'Maximal coupling' (`lem:coupling`), Lemma 'Ergodicity of the Gibbs sampler' (`lem:ergodic`) and Theorem `thm:dobrushin`; and, for Corollary 3.4, the proof of Theorem `thm:strong`. From [III] only Corollary 7.4, a restatement of [III, Corollary 'String tension' (`al:cor:tension`)] for the unique infinite-volume measure, is used.

2 The infinite lattice and the Wilson specification

2.1 Lattice, configurations, local functions

Convention 2.1 (The infinite lattice). Sites are $x \in \mathbb{Z}^d$, $d \geq 2$, with unit vectors $e_0, \dots, e_{d-1}$; direction 0 is Euclidean time and x_0 the time coordinate. Links $\ell = (x, \mu)$ and plaquettes $p = (x, \mu\nu)$, $\mu < \nu$, their sets $E := E(\mathbb{Z}^d)$ and $\mathcal{P} := \mathcal{P}(\mathbb{Z}^d)$, the four links $E(p)$ of a plaquette and the plaquette variable U_p are as in [F, Convention 'Lattice'] and [F, eq. `eq:plaquette`], with $\mathbb{Z}^d$ in place of the torus; $E(X) := \bigcup_{p \in X} E(p)$. For nearest neighbours y, z we write $U(y,z) := U_{(y,\mu)}$ if $z = y + e_\mu$ and $U(y,z) := U^{-1}_{(z,\mu)}$ if $y = z + e_\mu$ (parallel transport along the oriented edge), so that $U_p = U(x, x+e_\mu)\, U(x+e_\mu, x+e_\mu+e_\nu)\, U(x+e_\mu+e_\nu, x+e_\nu)\, U(x+e_\nu, x)$ for $p = (x, \mu\nu)$.

Lemma 2.2 (Geometry of links). *(a) Every link lies in exactly $2(d-1)$ plaquettes, two distinct links lie in at most one common plaquette, and two distinct plaquettes have at most one common link. Call distinct links* adjacent, *$\ell \sim \ell'$, if they lie in a common plaquette; every link has exactly $6(d-1)$ adjacent links. Let ϱ be the graph distance on E for $\sim$ and $\varrho(A,B) := \min\{\varrho(a,b) : a \in A, b \in B\}$.*

(b) For a link ℓ let $\|\ell\| := \max\{|y|_\infty, |z|_\infty\}$ for its endpoints y, z, and $t_\pm(\ell)$ the largest/smallest time coordinate of its endpoints. If $\ell \sim \ell'$ then every endpoint of ℓ' is at $|\cdot|_\infty$-distance at most 1 from every endpoint of ℓ. Consequently, for all links ℓ, ℓ',

$$\varrho(\ell, \ell') \;\geq\; \|\ell'\| - \|\ell\| \qquad \text{and} \qquad \varrho(\ell, \ell') \;\geq\; t_-(\ell') - t_+(\ell).$$

(c) Let $B_n := \{x : |x|_\infty \leq n\}$ and E_n the set of links with both endpoints in B_n. Every plaquette containing a link of E_n has its four corners in B_{n+1}, and every link outside E_n has $\|\ell\| \geq n+1$. Hence for $A \subseteq E_r$ and $n \geq r$, $\varrho(A, E \setminus E_n) \geq n + 1 - r$.

Proof. (a) The plaquettes containing $\ell = (x, \mu)$ are, for each $\nu \neq \mu$, the unit squares spanned by e_μ, e_ν with an edge on ℓ, one on each side of ℓ in direction ν: $2(d-1)$ squares. A unit square is determined by any two of its edges (two parallel edges span it; two edges meeting at a corner determine its plane and position), so two distinct links lie in at most one common plaquette, and two distinct plaquettes share at most one edge. The $2(d-1)$ plaquettes through ℓ therefore contribute 3 further links each, all distinct: $6(d-1)$ adjacent links. (b) All four corners of a plaquette are pairwise at $|\cdot|_\infty$-distance ≤ 1, and the endpoints of ℓ, ℓ' are corners of their common plaquette. Along a path $\ell = \ell_0 \sim \ell_1 \sim \cdots \sim \ell_k = \ell'$ this gives $\|\ell_{j+1}\| \leq \|\ell_j\| + 1$ and $t_+(\ell_{j+1}) \leq t_+(\ell_j) + 1$, hence $\|\ell'\| \leq \|\ell\| + k$ and $t_-(\ell') \leq t_+(\ell') \leq t_+(\ell) + k$. (c) The corners of a plaquette containing a link of E_n are within distance 1 of an endpoint of that link. A link not in E_n has an endpoint $z \notin B_n$, i.e. $|z|_\infty \geq n+1$. The last claim follows from (b), as $\|a\| \leq r$ for $a \in A$. □

Convention 2.3 (Configurations and local functions). $\Omega := G^E$ with the product topology, a compact metrisable space; its Borel σ-algebra $\mathcal{F}$ is the product σ-algebra (E is countable and G is second countable). For $A \subseteq E$, $\mathcal{F}_A := \sigma(U_\ell : \ell \in A)$. A function $F : \Omega \to \mathbb{C}$ *depends only on* A if $F(U) = F(U')$ whenever U, U' agree on A; for measurable F this holds iff F is $\mathcal{F}_A$-measurable. F is *local* if it is bounded, measurable and depends only on a finite set of links; $\mathcal{B}_{\mathrm{loc}}$ is the space of local functions and C_{loc} that of continuous local functions. For bounded F and $\ell \in E$, $\delta_\ell(F) := \sup\{|F(U) - F(U')| : U, U' \text{ agree off } \ell\}$ and $\|\delta(F)\|_1 := \sum_\ell \delta_\ell(F)$; if F depends only on a finite A, then $\delta_\ell(F) = 0$ for $\ell \notin A$, $\|\delta(F)\|_1 \le 2|A|\|F\|_\infty$, and $|F(U) - F(U')| \le \|\delta(F)\|_1$ for all U, U' (change the coordinates in A one at a time, as in [F, eq. `eq:interp`]). For a finite $\Delta \subseteq E$, $V \in G^\Delta$ and $\eta \in \Omega$, $V \vee \eta \in \Omega$ denotes the configuration equal to V on Δ and to η on $E \setminus \Delta$.

Lemma 2.4 (Measures on Ω). *(i) C_{loc} is dense in $C(\Omega)$ for the uniform norm.*

(ii) Two Borel probability measures on Ω which agree on C_{loc} (or on $\mathcal{B}_{\mathrm{loc}}$) are equal.

(iii) Every sequence of Borel probability measures on Ω has a weakly convergent subsequence.

(iv) For every Borel probability measure μ on Ω and every $A \subseteq E$, the bounded $\mathcal{F}_A$-measurable functions that depend only on finitely many links of A are dense in $L^2(\Omega, \mathcal{F}_A, \mu)$.

Proof. (i) C_{loc} is a subalgebra of $C(\Omega)$ containing the constants and closed under complex conjugation; it separates points, since configurations $U \neq U'$ differ at some link ℓ and a continuous function h on the compact Hausdorff space G with $h(U_\ell) \neq h(U'_\ell)$ exists (Urysohn). The Stone–Weierstrass theorem applies on the compact Hausdorff space Ω. (ii) By (i) the two measures agree on $C(\Omega)$, hence are equal by the uniqueness part of the Riesz representation theorem. (iii) is Prokhorov's theorem on the compact metrisable space Ω [Bil99, Section 5]. (iv) The union $\mathcal{A}_A$ of the σ-algebras $\mathcal{F}_{A'}$, $A' \subseteq A$ finite, is an algebra of sets generating $\mathcal{F}_A$. Let $\mathcal{S}$ be the class of $S \in \mathcal{F}_A$ such that for every $\epsilon > 0$ there is $S' \in \mathcal{A}_A$ with $\mu(S \triangle S') < \epsilon$. It contains $\mathcal{A}_A$, is closed under complements ($S^c \triangle S'^c = S \triangle S'$) and under countable unions (given $S_1, S_2, \ldots \in \mathcal{S}$, choose k with $\mu(\bigcup_i S_i \setminus \bigcup_{i \le k} S_i) < \epsilon/2$ and $S'_i \in \mathcal{A}_A$ with $\mu(S_i \triangle S'_i) < \epsilon/(2k)$ for $i \le k$; then $\bigcup_{i \le k} S'_i \in \mathcal{A}_A$ approximates $\bigcup_i S_i$ within ϵ). So $\mathcal{S} = \mathcal{F}_A$. Hence indicators of sets in $\mathcal{F}_A$, and therefore simple functions and all of $L^2(\mathcal{F}_A, \mu)$, are L^2-limits of linear combinations of indicators of sets in $\mathcal{A}_A$, which are of the stated type. □

2.2 The Wilson specification and Gibbs measures

Definition 2.5 (Wilson specification). Let $\beta \ge 0$. For a finite $\Delta \subseteq E$ let $\mathcal{P}_\Delta := \{p \in \mathcal{P} : E(p) \cap \Delta \neq \emptyset\}$ (a finite set) and $\partial\Delta := E(\mathcal{P}_\Delta) \setminus \Delta$. For $\eta \in \Omega$ and $V \in G^\Delta$ put

$$\rho_\Delta(V, \eta) := \prod_{p \in \mathcal{P}_\Delta} w_\beta((V \vee \eta)_p), \qquad Z_\Delta(\eta) := \int_{G^\Delta} \rho_\Delta(V, \eta)\, dV,$$

and for bounded measurable $F : \Omega \to \mathbb{C}$

$$\gamma_\Delta(F \,|\, \eta) := \frac{1}{Z_\Delta(\eta)} \int_{G^\Delta} F(V \vee \eta)\, \rho_\Delta(V, \eta)\, dV.$$

$\gamma_\Delta(\cdot \,|\, \eta)$ is the probability measure on Ω which resamples the links of Δ from the Wilson weight of the plaquettes touching Δ, the other links being fixed to η.

Lemma 2.6 (Properties of the specification). *Let $\Delta \subseteq E$ be finite and F bounded measurable.*

(a) $e^{-\beta|\mathcal{P}_\Delta|} \le Z_\Delta \le e^{\beta|\mathcal{P}_\Delta|}$; $\eta \mapsto \gamma_\Delta(F|\eta)$ is measurable, $|\gamma_\Delta(F|\eta)| \le \|F\|_\infty$, and if F depends only on A then $\gamma_\Delta(F|\cdot)$ depends only on $\partial\Delta \cup (A \setminus \Delta)$. In particular $\gamma_\Delta(F|\cdot)$ is $\mathcal{F}_{E \setminus \Delta}$-measurable, and local if F is.

(b) (Properness) *If H is bounded and $\mathcal{F}_{E\setminus\Delta}$-measurable, $\gamma_\Delta(HF|\eta) = H(\eta)\gamma_\Delta(F|\eta)$.*

(c) (Feller property) *If F is continuous, so is $\gamma_\Delta(F|\cdot)$.*

(d) (A criterion for the DLR equation) *Let κ be a probability measure on $G^{E\setminus\Delta}$ and $R' \geq 0$ a measurable function on $G^{E\setminus\Delta}$, and suppose that $m(dU) := \rho_\Delta(U_\Delta, U)\, R'(U_{E\setminus\Delta})\, dU_\Delta \otimes \kappa(dU_{E\setminus\Delta})$ is a probability measure on $\Omega = G^\Delta \times G^{E\setminus\Delta}$. Then $\int \gamma_\Delta(F|\eta)\, m(d\eta) = \int F\, dm$ for every bounded measurable F.*

(e) (Covariance) *For every gauge transformation $g \in G^{\mathbb{Z}^d}$, acting by $(g{\cdot}U)_{(x,\mu)} = g(x)U_{(x,\mu)}g(x+e_\mu)^{-1}$, one has $\gamma_\Delta(F\circ g\,|\,\eta) = \gamma_\Delta(F\,|\,g{\cdot}\eta)$, where $(F\circ g)(U) := F(g{\cdot}U)$. For every lattice symmetry s (Definition 2.7), $\gamma_{s\Delta}(F\,|\,s{\cdot}\eta) = \gamma_\Delta(F\circ s\,|\,\eta)$, where $(F\circ s)(U) := F(s\cdot U)$.*

Definition 2.7 (Lattice symmetries)**.** A *lattice symmetry* is a map $s(x) = Rx + a$ of $\mathbb{Z}^d$ with R a signed permutation matrix and $a \in \mathbb{Z}^d$. It permutes links and plaquettes (as unoriented edges and squares) and acts on Ω by $(s\cdot U)(y,z) := U(s^{-1}y, s^{-1}z)$ for nearest neighbours y, z (Convention 2.1); i.e. $(s\cdot U)_\ell = U_{s^{-1}\ell}^{\pm 1}$ with the sign -1 iff s^{-1} reverses the orientation of ℓ. Translations $\tau_a(x) := x - a$ act by $(\tau_a\cdot U)_{(x,\mu)} = U_{(x+a,\mu)}$. The time reflection is $\theta(x_0,\vec{x}) := (-x_0,\vec{x})$; then $(\theta\cdot U)_{(x,i)} = U_{(\theta x,i)}$ for $i \geq 1$ and $(\theta\cdot U)_{(x,0)} = U^{-1}_{(\theta x - e_0,0)}$, which is the map θ^* of [F, Section 'Reflection positivity'] transplanted to $\mathbb{Z}^d$.

Proof of Lemma 2.6. (a) The bounds on Z_Δ follow from $e^{-\beta} \leq w_\beta \leq e^\beta$ [F, eq. `eq:w`]. Measurability follows from Fubini's theorem, since $(V,\eta) \mapsto F(V\vee\eta)\rho_\Delta(V,\eta)$ is jointly measurable. The links of the plaquettes in $\mathcal{P}_\Delta$ lie in $\Delta \cup \partial\Delta$, so $\rho_\Delta(V,\eta)$ depends on η only through $\partial\Delta$, and $F(V\vee\eta)$ depends on η only through $A\setminus\Delta$. (b) $H(V\vee\eta) = H(\eta)$, since $V\vee\eta$ and η agree off Δ. (c) $(V,\eta) \mapsto V\vee\eta$ is continuous $G^\Delta\times\Omega\to\Omega$, and so is $U\mapsto U_p$; hence the integrands of $Z_\Delta(\eta)$ and of the numerator are continuous and bounded on $G^\Delta\times\Omega$, and the integrals are continuous in η by dominated convergence (sequences suffice, Ω being metrisable); and $Z_\Delta > 0$. (d) Write η for the variable in $G^{E\setminus\Delta}$ and V for the variable in G^Δ. By (a), $\gamma_\Delta(F|V\vee\eta) = \gamma_\Delta(F|\eta)$ does not depend on V, and $\int \rho_\Delta(V,\eta)dV = Z_\Delta(\eta)$. By Fubini,

$$\begin{aligned}\int \gamma_\Delta(F|\cdot)\, dm &= \int \kappa(d\eta)\, R'(\eta)\, \gamma_\Delta(F|\eta) \int \rho_\Delta(V,\eta)\, dV \\ &= \int \kappa(d\eta)\, R'(\eta) \int F(V\vee\eta)\, \rho_\Delta(V,\eta)\, dV = \int F\, dm.\end{aligned}$$

(e) Gauge: $g\cdot(V\vee\eta) = (g\cdot V)\vee(g\cdot\eta)$, where $(g\cdot V)_\ell = g(x)V_\ell g(x+e_\mu)^{-1}$ for $\ell = (x,\mu)\in\Delta$, and $(g\cdot U)_p = g(x)U_p g(x)^{-1}$ for $p = (x,\mu\nu)$ [F, Theorem `thm:basic`](ii), so $\rho_\Delta(V,\eta) = \rho_\Delta(g\cdot V, g\cdot\eta)$ because w_β is a class function. The map $V\mapsto g\cdot V$ is a product of left and right translations, hence preserves dV. Substituting $V' = g\cdot V$ in the integrals defining $\gamma_\Delta(F\circ g|\eta)$ gives $\gamma_\Delta(F|g\cdot\eta)$ (the same substitution shows $Z_\Delta(\eta) = Z_\Delta(g\cdot\eta)$). Lattice symmetries: $s\cdot(V\vee\eta) = V'\vee(s\cdot\eta)$ with $V'\in G^{s\Delta}$, $V'_{s\ell} = V_\ell^{\pm 1}$, a measure-preserving bijection $G^\Delta\to G^{s\Delta}$ (Haar measure is inversion invariant). By Convention 2.1, $(s\cdot U)_q$ is the holonomy of U around the boundary of the square $s^{-1}q$, from one of its corners and in one of its two orientations; holonomies of a closed path from different base points are conjugate, and reversal inverts them. As w_β is a class function invariant under inversion, $w_\beta((s\cdot U)_q) = w_\beta(U_{s^{-1}q})$. Since $q\in\mathcal{P}_{s\Delta}$ iff $s^{-1}q\in\mathcal{P}_\Delta$, $\rho_{s\Delta}(V',s\cdot\eta) = \rho_\Delta(V,\eta)$; substituting gives the claim. □

Definition 2.8 (Gibbs measures)**.** A *Gibbs measure* (for the Wilson specification at coupling β) is a Borel probability measure μ on Ω such that $\int \gamma_\Delta(F|\eta)\,\mu(d\eta) = \mu(F)$ for every finite $\Delta\subseteq E$ and every bounded measurable F (the DLR equations).

Lemma 2.9 (DLR and conditional expectations)**.** *If μ is a Gibbs measure, then for every finite Δ and bounded measurable F, $\gamma_\Delta(F|\cdot)$ is a version of $\mathbb{E}_\mu[F\,|\,\mathcal{F}_{E\setminus\Delta}]$. Every Gibbs measure is invariant under gauge transformations g with $g(x) = 1$ for all but finitely many x.*

Proof. $\gamma_\Delta(F|\cdot)$ is $\mathcal{F}_{E\setminus\Delta}$-measurable (Lemma 2.6(a)), and for $S \in \mathcal{F}_{E\setminus\Delta}$, by the DLR equation and properness, $\mu(\mathbf{1}_S F) = \mu(\gamma_\Delta(\mathbf{1}_S F|\cdot)) = \mu(\mathbf{1}_S \gamma_\Delta(F|\cdot))$. For the second claim let $g(x) = 1$ off a finite set X and let Δ be the (finite) set of links with an endpoint in X. Then $g\cdot\eta = \eta$ off Δ, so $\gamma_\Delta(F|g\cdot\eta) = \gamma_\Delta(F|\eta)$ by Lemma 2.6(a), and by the DLR equation and Lemma 2.6(e), $\mu(F\circ g) = \int \gamma_\Delta(F\circ g|\eta)\,\mu(d\eta) = \int \gamma_\Delta(F|g\cdot\eta)\,\mu(d\eta) = \int \gamma_\Delta(F|\eta)\,\mu(d\eta) = \mu(F)$. □

3 Dobrushin estimates for the Wilson specification

3.1 A sharp total-variation bound

Lemma 3.1 (Total variation under a bounded density ratio)**.** *Let π, π' be probability measures on a measurable space with $\pi' = h\,\pi$, and suppose $c\,e^{-\delta} \le h \le c\,e^{\delta}$ for some $c > 0$, $\delta \ge 0$. Then*

$$\|\pi - \pi'\|_{\mathrm{TV}} := \sup_S |\pi(S) - \pi'(S)| \;\le\; \tanh\frac{\delta}{2}.$$

Proof. Let $m := \operatorname{ess\,inf} h$ and $M := \operatorname{ess\,sup} h$ (with respect to π). Since $\int h\,d\pi = 1$, $m \le 1 \le M$, and $ce^{-\delta} \le m$, $M \le ce^{\delta}$. If $M = m$ then $h = 1$ a.e. and $\pi = \pi'$. Otherwise fix S and put $x := \pi(S)$, $y := \pi'(S) = \int_S h\,d\pi$. Then $y \le Mx$ and $1 - y = \int_{S^c} h\,d\pi \ge m(1-x)$, so $y - x \le \min\{(M-1)x, (1-m)(1-x)\} \le \frac{(M-1)(1-m)}{M-m}$, the value at the point $x = \frac{1-m}{M-m}$ where the increasing and the decreasing linear function coincide. Exchanging π, π' replaces h by $1/h$ and (m, M) by $(1/M, 1/m)$, which gives the same bound for $x - y$. The function $f(m, M) := \frac{(M-1)(1-m)}{M-m}$ is non-decreasing in M and non-increasing in m on $\{m \le 1 \le M,\ m < M\}$ (its partial derivatives are $\frac{(1-m)^2}{(M-m)^2}$ and $-\frac{(M-1)^2}{(M-m)^2}$), so with $ce^{-\delta} \le m \le 1 \le M \le ce^{\delta}$,

$$f(m, M) \le f(ce^{-\delta}, ce^{\delta}) = \frac{2\cosh\delta - c - c^{-1}}{2\sinh\delta} \le \frac{2\cosh\delta - 2}{2\sinh\delta} = \tanh\frac{\delta}{2},$$

using $c + c^{-1} \ge 2$ (if $\delta = 0$ then $m = M$ and there is nothing to prove). □

Remark 3.2. The bound is sharp: on a two-point space with $\pi = (\frac{1}{1+e^{\delta}}, \frac{e^{\delta}}{1+e^{\delta}})$ and $h = (e^{\delta}, e^{-\delta})$ (so that $\int h\,d\pi = 1$ and $c = 1$) one has $\pi' = (\frac{e^{\delta}}{1+e^{\delta}}, \frac{1}{1+e^{\delta}})$ and $\|\pi - \pi'\|_{\mathrm{TV}} = \frac{e^{\delta}-1}{e^{\delta}+1} = \tanh\frac{\delta}{2}$. In [F, Lemma `lem:dobrushinbound`] the same density ratio is bounded by $e^{\pm 4\beta}$ (the normalisations being estimated separately) and the total variation by $e^{4\beta} - 1 \approx 4\beta$; Lemma 3.3 below gives $\tanh\beta \approx \beta$. This is the source of the factor ≈ 4 between $\beta_D(d)$ and $\beta_*(d)$.

3.2 Dobrushin coefficients

We use the notions of [F, Definition 'Finite spin system' (`def:spin`)]: a finite set V, the compact metric space $S = G$ with Haar measure, a continuous positive density ρ on G^V, the resulting measure μ, the single-site conditional distributions $\mu_x(\cdot|\omega)$, the Dobrushin matrix $C = (C_{xy})_{x,y\in V}$ and its maximal row sum α.

Lemma 3.3 (Dobrushin coefficients of plaquette systems)**.** *Let $V \subseteq E$ be finite, $\Phi \subseteq \mathcal{P}$ finite, $\xi \in \Omega$, and $\rho(U) := \prod_{p\in\Phi} w_\beta((U \vee \xi)_p)$ for $U \in G^V$. Then the Dobrushin matrix of (V, G, dU, ρ) satisfies, for $x \ne y$ in V: $C_{xy} = 0$ unless x and y lie in a common plaquette of Φ, and $C_{xy} \le \tanh\beta$ always. Hence its row sums are at most $\alpha_*(\beta) = 6(d-1)\tanh\beta$, and $C_{xy} = 0$ unless $x \sim y$. The same holds for the Wilson measure of a periodic lattice with all $L_\mu \ge 3$ ($V = E(\Lambda)$, $\Phi = \mathcal{P}(\Lambda)$).*

Proof. The single-site conditional distribution at x has density proportional to

$$s \;\longmapsto\; \prod_{p\in\Phi,\, p\ni x} w_\beta(((\omega^{x\leftarrow s}) \vee \xi)_p)$$

with respect to Haar measure. Let ω, ω' agree off $y \neq x$. Factors of plaquettes not containing y coincide, and by Lemma 2.2(a) at most one plaquette p contains both x and y; if there is none, $C_{xy} = 0$. Otherwise the two densities are $e^{\varphi(s)}/Z$ and $e^{\varphi'(s)}/Z'$ with $\varphi(s) - \varphi'(s) = \log w_\beta(\cdot) - \log w_\beta(\cdot)$ evaluated on the two versions of U_p, a difference of two numbers in $[-\beta, \beta]$ [F, eq. `eq:w`]. Their ratio is $(Z/Z')e^{\varphi'(s)-\varphi(s)} \in [ce^{-2\beta}, ce^{2\beta}]$ with $c = Z/Z'$, and Lemma 3.1 with $\delta = 2\beta$ gives $\|\mu_x(\cdot|\omega) - \mu_x(\cdot|\omega')\|_{\mathrm{TV}} \leq \tanh\beta$. A link has $6(d-1)$ adjacent links (Lemma 2.2(a)), whence the row sums. On a periodic lattice with all $L_\mu \geq 3$, two distinct links lie in at most one common plaquette and each link has $6(d-1)$ adjacent links ([F, Section 'Application to the Wilson measure']), and the same argument applies. □

Corollary 3.4 (Improved strong-coupling gap in finite volume)**.** *Let $d \geq 2$ and $0 < \beta < \beta_*(d)$. Then [F, Theorem 'Strong-coupling clustering and mass gap, uniform in the volume' (`thm:strong`)] holds with $\alpha(\beta)$ replaced by $\alpha_*(\beta)$. In particular $m_{\Lambda_s}(\beta) \geq \log(1/\alpha_*(\beta))$ for every spatial lattice Λ_s with all $L_i \geq 3$.*

Proof. The proof of [F, Theorem `thm:strong`] uses [F, Lemma `lem:dobrushinbound`] only through its two conclusions: $C_{\ell\ell'} = 0$ unless $\ell \sim \ell'$, and the row sums are at most $\alpha(\beta) < 1$. By Lemma 3.3 (periodic case) both hold with $\alpha_*(\beta) < 1$ in place of $\alpha(\beta)$; [F, Theorem `thm:dobrushin`] and the rest of that proof (via [F, Corollary `cor:gapfromcorr`]) then apply word for word. □

3.3 Influence of the boundary

Lemma 3.5 (Boundary influence)**.** *Let (V, S, σ, ρ) be a finite spin system as in [F, Definition `def:spin`] with Dobrushin matrix C, maximal row sum $\alpha < 1$, and $D := \sum_{n\geq 1} C^n$. Let $V = V' \sqcup B$ with $V' \neq \emptyset$, and for $\eta \in S^B$ let μ^η be the probability measure on $S^{V'}$ with density proportional to $\rho(\cdot, \eta)$. Let $\eta, \eta' \in S^B$ and $B_1 := \{y \in B : \eta_y \neq \eta'_y\}$. If $f : S^{V'} \to \mathbb{C}$ is bounded, measurable and depends only on $A \subseteq V'$, then*

$$|\mu^\eta(f) - \mu^{\eta'}(f)| \leq \sum_{x\in A} \delta_x(f) \sum_{y\in B_1} D_{xy}.$$

If moreover $C_{xy} = 0$ unless $x \sim y$ for a graph structure on V with graph distance ϱ_V, the right-hand side is at most $\|\delta(f)\|_1\, \alpha^{\varrho_V(A,B_1)}/(1-\alpha)$ (and it is 0 if $B_1 = \emptyset$).

Proof. This is Step 1 of the proof of [F, Theorem `thm:dobrushin`], keeping track of which boundary sites disagree. Let $n := |V'|$. On a common probability space construct a Markov chain $(\omega^k, \omega'^k)_{k\geq 0}$ on $S^{V'} \times S^{V'}$: $\omega^0 = \omega'^0$ is a fixed configuration; at step $k+1$ a site $x_{k+1} \in V'$ is chosen uniformly and independently of the past, all other coordinates are kept, and the pair of new values at $x := x_{k+1}$ is drawn from the maximal coupling ([F, Lemma `lem:coupling`]) of $\mu_x(\cdot|(\omega^k, \eta))$ and $\mu_x(\cdot|(\omega'^k, \eta'))$ (these have continuous densities with respect to σ). Marginally, (ω^k) and (ω'^k) are the random-scan Gibbs samplers for μ^η and $\mu^{\eta'}$. Let $r_k(x) := \mathbb{P}(\omega^k_x \neq \omega'^k_x)$ for $x \in V'$, and $\tilde r_k(y) := r_k(y)$ for $y \in V'$, $\tilde r_k(y) := \mathbf{1}_{B_1}(y)$ for $y \in B$. The configurations (ω^k, η) and (ω'^k, η') differ exactly on $\{y \in V' : \omega^k_y \neq \omega'^k_y\} \cup B_1$, so by [F, eq. `eq:interp`] the conditional probability that the new values at x differ is at most $\sum_y C_{xy}\mathbf{1}[\text{disagreement at } y]$; taking expectations,

$$r_{k+1}(x) \leq \Big(1 - \frac{1}{n}\Big) r_k(x) + \frac{1}{n} \sum_{y\in V} C_{xy}\, \tilde r_k(y) \qquad (x \in V').$$

Let C' be the restriction of C to $V' \times V'$ and C'' the restriction to $V' \times B_1$, and put $r^* := \sum_{m\geq 0} (C')^m C'' \mathbf{1}_{B_1} \in [0, \infty)^{V'}$ (convergent, since the row sums of C' are $\leq \alpha < 1$). Then $r^* = C' r^* + C''\mathbf{1}_{B_1}$, and by induction $r_k \leq r^*$ componentwise: $r_0 = 0$, and if $r_k \leq r^*$

then, as $C \geq 0$, $r_{k+1} \leq (1-\frac{1}{n})r^* + \frac{1}{n}(C'r^* + C''\mathbf{1}_{B_1}) = r^*$. Since $((C')^m C'')_{xy}$ is the sum of the products of the entries of C along the paths $x \to \cdots \to y$ of length $m+1$ whose intermediate vertices lie in V', it is at most $(C^{m+1})_{xy}$; hence $r^*_x \leq \sum_{y\in B_1} D_{xy}$. By [F, eq. `eq:interp`], $|\mathbb{E}f(\omega^k) - \mathbb{E}f(\omega'^k)| \leq \sum_{x\in A}\delta_x(f) r_k(x) \leq \sum_{x\in A}\delta_x(f)\sum_{y\in B_1} D_{xy}$, and by [F, Lemma `lem:ergodic`] the left side converges to $|\mu^\eta(f) - \mu^{\eta'}(f)|$ as $k\to\infty$. For the last claim, $(C^m)_{xy} = 0$ unless $\varrho_V(x,y) \leq m$, and $\sum_y (C^m)_{xy} \leq \alpha^m$; so for $x \in A$, $\sum_{y\in B_1} D_{xy} \leq \sum_{m \geq \varrho_V(x,B_1)} \alpha^m \leq \alpha^{\varrho_V(A,B_1)}/(1-\alpha)$, using $\varrho_V(x,B_1)\geq 1$. □

Lemma 3.6 (Boundary influence for the Wilson specification)**.** *Let $0 < \beta < \beta_*(d)$ and $\alpha = \alpha_*(\beta)$. Let $\Delta \subseteq E$ be finite, F bounded measurable depending only on a finite $A \subseteq E$, and $\eta, \eta' \in \Omega$ such that $\eta = \eta'$ on $A \setminus \Delta$. With $B_1 := \{\ell \in \partial\Delta : \eta_\ell \neq \eta'_\ell\}$,*

$$|\gamma_\Delta(F|\eta) - \gamma_\Delta(F|\eta')| \leq \frac{\|\delta(F)\|_1}{1-\alpha}\,\alpha^{\varrho(A\cap\Delta, B_1)}.$$

In particular, if $A \subseteq E_r$ and $n \geq r$, then $\sup_{\eta,\eta'\in\Omega} |\gamma_{E_n}(F|\eta) - \gamma_{E_n}(F|\eta')| \leq \|\delta(F)\|_1 \alpha^{n+1-r}/(1-\alpha)$.

Proof. Consider the finite spin system with $V := \Delta \cup \partial\Delta$, $\rho(U) := \prod_{p\in\mathcal{P}_\Delta} w_\beta(U_p)$ (all links of these plaquettes lie in V), $V' := \Delta$ and $B := \partial\Delta$. For $\zeta \in \Omega$, $\gamma_\Delta(\cdot|\zeta)$ restricted to the links of Δ is $\mu^{\zeta|_{\partial\Delta}}$. Put $f(V) := F(V \vee \eta) = F(V \vee \eta')$ for $V \in G^\Delta$ (equal, since F depends only on A and $\eta = \eta'$ on $A\setminus\Delta$); f depends only on $A \cap \Delta$ and $\delta_x(f) \leq \delta_x(F)$. Then $\gamma_\Delta(F|\eta) = \mu^{\eta|_{\partial\Delta}}(f)$ and $\gamma_\Delta(F|\eta') = \mu^{\eta'|_{\partial\Delta}}(f)$. By Lemma 3.3 (with $\Phi = \mathcal{P}_\Delta$) this system has $C_{xy} = 0$ unless $x \sim y$ and row sums $\leq \alpha$, and the graph distance ϱ_V of the subgraph induced on V is at least ϱ. Lemma 3.5 gives the first claim. For the second, $A \setminus E_n = \emptyset$, and $\partial E_n \subseteq E \setminus E_n$, so $\varrho(A, \partial E_n) \geq n+1-r$ by Lemma 2.2(c). □

4 The infinite-volume Gibbs measure

4.1 Finite-volume measures satisfy the DLR equation in boxes

Definition 4.1 (Finite-volume measures and limit points)**.** Let $\beta \geq 0$.

(a) *Periodic.* For a periodic lattice $\Lambda = \prod_\mu \mathbb{Z}/L_\mu\mathbb{Z}$ (all $L_\mu \geq 2$) let $\pi : \mathbb{Z}^d \to \Lambda$ be the reduction map, also applied to links, $\pi(x,\mu) := (\pi x, \mu)$, and plaquettes. Let $\iota_\Lambda : \Omega_\Lambda \to \Omega$, $(\iota_\Lambda U)_\ell := U_{\pi\ell}$, and $\tilde{\mu}_{\Lambda,\beta} := (\iota_\Lambda)_* \mu_{\Lambda,\beta}$, where $\mu_{\Lambda,\beta}$ is the Wilson measure of [F, Definition 'Wilson measure' (`def:wilson`)].

(b) *Free boundary.* For $n' \geq 1$ and $\xi \in \Omega$, $\mu^{\mathrm{free}}_{n',\xi}$ is the probability measure with density proportional to $\prod_{p\in\mathcal{P}(B_{n'})} w_\beta(U_p)$ with respect to $dU_{E_{n'}} \otimes \delta_{\xi|_{E\setminus E_{n'}}}$, where $\mathcal{P}(B_{n'})$ is the set of plaquettes with all four corners in $B_{n'}$.

(c) *Fixed boundary.* For $n' \geq 0$ and $\eta \in \Omega$, the measure $\gamma_{E_{n'}}(\cdot\,|\,\eta)$.

A Borel probability measure μ on Ω is an *infinite-volume limit point* (at coupling β) if $\tilde{\mu}_{\Lambda_k,\beta} \to \mu$ weakly along some sequence of periodic lattices Λ_k with $\min_\mu L^{(k)}_\mu \to \infty$ (this is the notion of [III, Definition 'Periodisation and infinite-volume limit points' (`al:def:limitpoints`)]).

Lemma 4.2 (DLR equation in boxes)**.** *Let $n \geq 0$, $\Delta \subseteq E_n$, and let F be bounded, measurable and dependent only on E_n. Then $\int \gamma_\Delta(F|\eta)\, m(d\eta) = \int F\, dm$ for each of the following measures m:*

(a) $m = \tilde{\mu}_{\Lambda,\beta}$, for every periodic lattice Λ with all $L_\mu \geq 2n+3$;

(b) $m = \mu^{\mathrm{free}}_{n',\xi}$, for every $n' \geq n+1$ and $\xi \in \Omega$;

(c) $m = \gamma_{E_{n'}}(\cdot|\eta)$*, for every* $n' \geq n$ *and* $\eta \in \Omega$*.*

In cases (b) and (c) the identity holds for every bounded measurable F*.*

Proof. (c) Put $\Delta' := E_{n'} \supseteq \Delta$. The density of m with respect to $dU_{\Delta'} \otimes \delta_{\eta|_{E\setminus\Delta'}}$ is

$$Z_{\Delta'}(\eta)^{-1} \prod_{p\in\mathcal{P}_{\Delta'}} w_\beta((U_{\Delta'} \vee \eta)_p) = \rho_\Delta(U_\Delta, U)\, R'(U_{E\setminus\Delta}) \qquad \text{on the support of } m,$$

where R' is the (measurable) product over $\mathcal{P}_{\Delta'} \setminus \mathcal{P}_\Delta$ divided by $Z_{\Delta'}(\eta)$; this uses $\mathcal{P}_\Delta \subseteq \mathcal{P}_{\Delta'}$ and that the plaquettes of $\mathcal{P}_{\Delta'} \setminus \mathcal{P}_\Delta$ contain no link of Δ. So m has the form of Lemma 2.6(d) with $\kappa := dU_{\Delta'\setminus\Delta} \otimes \delta_{\eta|_{E\setminus\Delta'}}$. (b) By Lemma 2.2(c) every plaquette of $\mathcal{P}_\Delta$ has its corners in $B_{n+1} \subseteq B_{n'}$, so $\mathcal{P}_\Delta \subseteq \mathcal{P}(B_{n'})$ and the factors of $\mathcal{P}(B_{n'}) \setminus \mathcal{P}_\Delta$ do not involve the links of Δ; conclude as in (c) with $\kappa := dU_{E_{n'}\setminus\Delta} \otimes \delta_{\xi|_{E\setminus E_{n'}}}$. (a) Two points of B_{n+1} differ by a vector with entries of absolute value $\leq 2n+2 < L_\mu$, so π is injective on B_{n+1}, hence on E_{n+1} and on the plaquettes with corners in B_{n+1}, among them those of $\mathcal{P}_\Delta$. For $\ell = (x,\mu) \in \Delta$ the plaquettes of Λ containing $\pi\ell$ are those based at πx and at $\pi x - e_\nu$ in the planes $\{\mu,\nu\}$, $\nu \neq \mu$, i.e. the images of the plaquettes of $\mathbb{Z}^d$ containing ℓ. Hence π maps $\mathcal{P}_\Delta$ bijectively onto $\mathcal{P}^\Lambda_{\pi\Delta} := \{q \in \mathcal{P}(\Lambda) : E(q) \cap \pi\Delta \neq \emptyset\}$. Let $\gamma^\Lambda_{\pi\Delta}$ be the torus analogue of Definition 2.5. The torus Wilson density factorises as $\prod_{q\in\mathcal{P}^\Lambda_{\pi\Delta}} w_\beta(U_q)$ times a factor not involving $U_{\pi\Delta}$, so by the torus version of Lemma 2.6(d) (same proof), $\mu_{\Lambda,\beta}(\gamma^\Lambda_{\pi\Delta}(H|\cdot)) = \mu_{\Lambda,\beta}(H)$ for all bounded measurable H on Ω_Λ. We claim that

$$\gamma_\Delta(F \,|\, \iota_\Lambda U') = \gamma^\Lambda_{\pi\Delta}(F \circ \iota_\Lambda \,|\, U') \qquad (U' \in \Omega_\Lambda),$$

which gives

$$\tilde\mu_{\Lambda,\beta}(\gamma_\Delta(F|\cdot)) = \mu_{\Lambda,\beta}(\gamma^\Lambda_{\pi\Delta}(F\circ\iota_\Lambda|\cdot)) = \mu_{\Lambda,\beta}(F\circ\iota_\Lambda) = \tilde\mu_{\Lambda,\beta}(F).$$

To prove the claim identify $V \in G^\Delta$ with $V' \in G^{\pi\Delta}$, $V'_{\pi\ell} := V_\ell$, a measure-preserving bijection. For $\ell \in E_{n+1}$ the configurations $V \vee \iota_\Lambda U'$ and $\iota_\Lambda(V' \vee U')$ agree at ℓ: both equal V_ℓ if $\ell \in \Delta$, and if $\ell \notin \Delta$ then $\pi\ell \notin \pi\Delta$ (injectivity on E_{n+1}) and both equal $U'_{\pi\ell}$. As F depends only on E_n and the links of the plaquettes of $\mathcal{P}_\Delta$ lie in E_{n+1}, $F(V \vee \iota_\Lambda U') = F(\iota_\Lambda(V' \vee U'))$ and $(V \vee \iota_\Lambda U')_p = (V' \vee U')_{\pi p}$ for $p \in \mathcal{P}_\Delta$; so the integrands defining the two sides agree, and so do the normalisations. □

4.2 Limit points at every coupling

Proposition 4.3 (Limit points are Gibbs measures)**.** *Let* $\beta \geq 0$ *and let* Λ_k *be periodic lattices with* $\min_\mu L^{(k)}_\mu \to \infty$*.*

(a) $(\tilde\mu_{\Lambda_k,\beta})_k$ *has weakly convergent subsequences, and every limit point is a Gibbs measure. In particular Gibbs measures exist for every* $\beta \geq 0$*.*

(b) Every limit point is invariant under all translations τ_a*,* $a \in \mathbb{Z}^d$*. If all* $L^{(k)}_0$ *are even, it is also invariant under the time reflection* θ*.*

Proof. (a) Limit points exist by Lemma 2.4(iii). Let $\tilde\mu_{\Lambda_{k_j},\beta} \to \mu$ weakly, let Δ be finite and $F \in C_{\mathrm{loc}}$, and choose n such that $\Delta \subseteq E_n$ and F depends only on E_n. By Lemma 2.6(c), $\gamma_\Delta(F|\cdot)$ is continuous, and by Lemma 4.2(a), $\tilde\mu_{\Lambda_{k_j},\beta}(\gamma_\Delta(F|\cdot)) = \tilde\mu_{\Lambda_{k_j},\beta}(F)$ for j large. Letting $j \to \infty$, $\mu(\gamma_\Delta(F|\cdot)) = \mu(F)$. Thus the probability measures $F \mapsto \mu(\gamma_\Delta(F|\cdot))$ and μ agree on C_{loc}, hence are equal (Lemma 2.4(ii)): μ satisfies the DLR equation for Δ. (b) For $a \in \mathbb{Z}^d$ let $t_{\pi a}$ be the translation of Ω_Λ with $(t_{\pi a}U')_{(y,\mu)} = U'_{(y+\pi a,\mu)}$. Then $\tau_a \cdot \iota_\Lambda U' = \iota_\Lambda(t_{\pi a}U')$,

and $\mu_{\Lambda,\beta}$ is invariant under $t_{\pi a}$ [F, Theorem `thm:basic`](iv); so $\tilde{\mu}_{\Lambda,\beta}$ is τ_a-invariant. If L_0 is even, the reflection θ^*_Λ of [F, Section 'Reflection positivity'] satisfies $\theta \cdot \iota_\Lambda U' = \iota_\Lambda(\theta^*_\Lambda U')$: for spatial links, $(\theta \cdot \iota_\Lambda U')_{(x,i)} = U'_{(\pi\theta x,i)} = U'_{(\theta_\Lambda \pi x,i)}$, and for temporal links $(\theta \cdot \iota_\Lambda U')_{(x,0)} = U'^{-1}_{(\pi(\theta x - e_0),0)} = U'^{-1}_{(\theta_\Lambda \pi x - e_0,0)}$, where θ_Λ is the time reflection of Λ; and $\mu_{\Lambda,\beta}$ is θ^*_Λ-invariant [F, Lemma `lem:reflsym`]. So $\tilde{\mu}_{\Lambda,\beta}$ is θ-invariant. Invariance under a homeomorphism T of Ω passes to weak limits: $\mu(F \circ T) = \lim_j \tilde{\mu}_{\Lambda_{k_j},\beta}(F \circ T) = \lim_j \tilde{\mu}_{\Lambda_{k_j},\beta}(F) = \mu(F)$ for continuous F. □

4.3 Uniqueness and convergence at strong coupling

Theorem 4.4 (Uniqueness and convergence)**.** *Let $d \geq 2$, $0 < \beta < \beta_*(d)$ and $\alpha := \alpha_*(\beta)$.*

(i) There is exactly one Gibbs measure, denoted μ_β.

(ii) Let F be bounded, measurable and dependent only on E_r, let $n \geq r$, and let m be a probability measure on Ω with $\int \gamma_{E_n}(F|\eta)\, m(d\eta) = \int F\, dm$. Then

$$|m(F) - \mu_\beta(F)| \leq \frac{\|\delta(F)\|_1}{1-\alpha}\, \alpha^{n+1-r}.$$

By Lemma 4.2 this applies to $\tilde{\mu}_{\Lambda,\beta}$ if all $L_\mu \geq 2n+3$, to $\mu^{\mathrm{free}}_{n',\xi}$ if $n' \geq n+1$, and to $\gamma_{E_{n'}}(\cdot|\eta)$ if $n' \geq n$.

(iii) If Λ_k are periodic lattices with $\min_\mu L^{(k)}_\mu \to \infty$, then $\tilde{\mu}_{\Lambda_k,\beta}(F) \to \mu_\beta(F)$ for every $F \in \mathcal{B}_{\mathrm{loc}}$; in particular $\tilde{\mu}_{\Lambda_k,\beta} \to \mu_\beta$ weakly. The same holds for free and fixed boundary conditions as $n' \to \infty$, uniformly in ξ, resp. η.

Proof. Let m_1, m_2 be probability measures satisfying the hypothesis of (ii) for the same F and n. Then

$$|m_1(F) - m_2(F)| = \Big| \iint \big[\gamma_{E_n}(F|\eta) - \gamma_{E_n}(F|\eta')\big]\, m_1(d\eta)\, m_2(d\eta') \Big| \leq \frac{\|\delta(F)\|_1}{1-\alpha}\, \alpha^{n+1-r}$$

by Lemma 3.6. (i) A Gibbs measure exists by Proposition 4.3(a). If μ, μ' are Gibbs measures, they satisfy the hypothesis of (ii) for every local F and every $n \geq r$; letting $n \to \infty$ gives $\mu(F) = \mu'(F)$ for all $F \in \mathcal{B}_{\mathrm{loc}}$, so $\mu = \mu'$ (Lemma 2.4(ii)). (ii) is the displayed estimate with $m_1 = m$, $m_2 = \mu_\beta$. (iii) For F depending only on E_r and k large, Lemma 4.2(a) applies with $n_k := \lfloor (\min_\mu L^{(k)}_\mu - 3)/2 \rfloor \geq r$, and (ii) gives $|\tilde{\mu}_{\Lambda_k,\beta}(F) - \mu_\beta(F)| \leq \|\delta(F)\|_1 \alpha^{n_k+1-r}/(1-\alpha) \to 0$. For $F \in C(\Omega)$ and $\epsilon > 0$ pick $F' \in C_{\mathrm{loc}}$ with $\|F - F'\|_\infty < \epsilon$ (Lemma 2.4(i)); then $\limsup_k |\tilde{\mu}_{\Lambda_k,\beta}(F) - \mu_\beta(F)| \leq 2\epsilon$. Free and fixed boundary conditions are handled in the same way with Lemma 4.2(b),(c). □

Remark 4.5 (Rate). For a periodic lattice with all sides $\geq L \geq 3$ and F depending only on E_r, (ii) gives $|\langle F\rangle_{\Lambda,\beta} - \mu_\beta(F)| \leq \|\delta(F)\|_1\, \alpha^{\lfloor (L-1)/2 \rfloor - r}/(1-\alpha)$, where $\langle F\rangle_{\Lambda,\beta} := \tilde{\mu}_{\Lambda,\beta}(F)$; the bound is trivially true (since $|\tilde{\mu}(F) - \mu(F)| \leq \operatorname{osc} F \leq \|\delta(F)\|_1$) when the exponent is not positive. The finite-volume corrections to local expectations are thus exponentially small in the linear size, at rate at least $\frac{1}{2}\log(1/\alpha_*(\beta))$.

Theorem 4.6 (Symmetries)**.** *Let $0 < \beta < \beta_*(d)$. The Gibbs measure μ_β is invariant under every lattice symmetry (Definition 2.7) and under every gauge transformation $g \in G^{\mathbb{Z}^d}$, including those with infinite support.*

Proof. Let T be either $U \mapsto s \cdot U$ or $U \mapsto g \cdot U$, and $\mu'(F) := \mu_\beta(F \circ T)$, where $(F \circ T)(U) := F(T(U))$. For the gauge transformation, Lemma 2.6(e) gives $\int \gamma_\Delta(F|\eta)\, \mu'(d\eta) = \int \gamma_\Delta(F|g \cdot \eta)\, \mu_\beta(d\eta) = \int \gamma_\Delta(F \circ g|\eta)\, \mu_\beta(d\eta) = \mu_\beta(F \circ g) = \mu'(F)$. For the lattice symmetry, Lemma 2.6(e) applied to $s^{-1}\Delta$ gives $\gamma_\Delta(F|s \cdot \eta) = \gamma_{s^{-1}\Delta}(F \circ s|\eta)$, and the same computation applies. So μ' is a Gibbs measure, and $\mu' = \mu_\beta$ by Theorem 4.4(i). □

Proposition 4.7 (Gauge-variant variables: forests)**.** *Let $0 < \beta < \beta_*(d)$ and let $A \subseteq E$ be a finite set of links containing no cycle (the graph on $\mathbb{Z}^d$ with edge set A is a forest). Then under μ_β the variables $(U_\ell)_{\ell\in A}$ are independent and Haar distributed. In particular $\mu_\beta(D^r_{ab}(U_\ell)) = 0$ for every link ℓ and every irreducible $r \neq 0$ (Elitzur's theorem in infinite volume).*

Proof. Let X be the set of endpoints of the links of A, and m the law of $U_A := (U_\ell)_{\ell\in A}$. By Theorem 4.6, for every $g \in G^X$ (extended by 1 off X) m is invariant under $u \mapsto (g \cdot u)_A$, where $(g \cdot u)_{(x,\mu)} = g(x)u_{(x,\mu)}g(x + e_\mu)^{-1}$. Averaging over g distributed according to the Haar measure dg of G^X, $m(F) = \int m(du) \int F((g \cdot u)_A)\, dg$ for bounded measurable F on G^A. It therefore suffices to show that for every fixed $u \in G^A$ the law of $(g \cdot u)_A$ under dg is the Haar measure of G^A. We argue by induction on $|A|$, the case $A = \emptyset$ being trivial. A finite forest with an edge has a vertex $v \in X$ lying on exactly one link $\ell_v \in A$. The coordinates $(g\cdot u)_\ell$, $\ell \in A\setminus\{\ell_v\}$, do not involve $g(v)$, and by the induction hypothesis (applied to the forest $A \setminus \{\ell_v\}$, whose endpoints lie in $X \setminus \{v\}$) their law under dg is the Haar measure of $G^{A\setminus\{\ell_v\}}$. Conditionally on $(g(x))_{x\neq v}$, $(g \cdot u)_{\ell_v}$ is $g(y)u_{\ell_v}g(v)^{-1}$ or $g(v)u_{\ell_v}g(y)^{-1}$ (y the other endpoint) with $g(v)$ Haar distributed, hence Haar distributed by right or left invariance, whatever the conditioning. So the joint law is the product Haar measure. The last claim follows from [F, Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`)](ii) with the trivial representation. □

5 Clustering and the Markov property

Theorem 5.1 (Exponential clustering)**.** *Let $0 < \beta < \beta_*(d)$ and $\alpha := \alpha_*(\beta)$. For bounded measurable f, g depending only on disjoint finite sets of links A, B,*

$$|\mathrm{Cov}_{\mu_\beta}(f, g)| \leq \frac{\|g\|_\infty \|\delta(f)\|_1}{1 - \alpha}\, \alpha^{\varrho(A,B)}.$$

Proof. Let r be such that $A \cup B \subseteq E_r$, let $n \geq r$, $\Delta := E_n$, and write $\gamma(\cdot) := \gamma_\Delta(\cdot|\eta)$ for fixed η. By the DLR equation applied to fg, f and g,

$$\mathrm{Cov}_{\mu_\beta}(f, g) = \int \mathrm{Cov}_{\gamma_\Delta(\cdot|\eta)}(f, g)\, \mu_\beta(d\eta) + \mathrm{Cov}_{\mu_\beta}(\gamma_\Delta(f|\cdot), \gamma_\Delta(g|\cdot)).$$

For fixed η, $\gamma_\Delta(\cdot|\eta)$ is the finite spin system on $V = \Delta$ with density $\rho_\Delta(\cdot, \eta)$, and $f(\cdot \vee \eta)$, $g(\cdot \vee \eta)$ depend only on A, B. By Lemma 3.3 (with $V = \Delta$, $\Phi = \mathcal{P}_\Delta$, $\xi = \eta$) its Dobrushin matrix vanishes off adjacent pairs and has row sums $\leq \alpha$, and the graph distance in Δ is at least ϱ; so [F, Theorem `thm:dobrushin`] bounds the first term by $\|g\|_\infty\|\delta(f)\|_1\alpha^{\varrho(A,B)}/(1 - \alpha)$. For the second term, $|\mathrm{Cov}_\mu(F_1, F_2)| = |\mu((F_1 - \mu F_1)F_2)| \leq \|F_2\|_\infty \operatorname{osc} F_1$ with $\operatorname{osc} F_1 := \sup_{\eta,\eta'} |F_1(\eta) - F_1(\eta')|$; with $F_1 = \gamma_\Delta(f|\cdot)$ and $F_2 = \gamma_\Delta(g|\cdot)$, Lemma 3.6 bounds it by $\|g\|_\infty\|\delta(f)\|_1\alpha^{n+1-r}/(1 - \alpha)$, which tends to 0 as $n \to \infty$. □

Definition 5.2 (Half-spaces and time slices)**.** Let E_+ (E_-) be the set of links with both endpoints in $\{x_0 \geq 0\}$ (in $\{x_0 \leq 0\}$), and $E_0 := E_+ \cap E_-$, the set of spatial links (x, i), $1 \leq i \leq d - 1$, with $x_0 = 0$. More generally $\mathcal{F}_{\leq t}$ ($\mathcal{F}_{\geq t}$) is the σ-algebra of the links with both endpoints in $\{x_0 \leq t\}$ (in $\{x_0 \geq t\}$), so that $\mathcal{F}_{\leq 0} = \mathcal{F}_{E_-}$ and $\mathcal{F}_{\geq 0} = \mathcal{F}_{E_+}$. Let E_s be the set of links of $\mathbb{Z}^{d-1}$, $\Omega_s := G^{E_s}$ (compact metrisable), and for $t \in \mathbb{Z}$ let $U^{(t)} \in \Omega_s$ be the configuration of the spatial links in the slice $x_0 = t$: $U^{(t)}_{(\vec{x},i)} := U_{((t,\vec{x}),i)}$. Thus $\sigma(U^{(0)}) = \mathcal{F}_{E_0}$.

Theorem 5.3 (Global Markov property)**.** *Let $0 < \beta < \beta_*(d)$. For every $F \in L^2(\Omega, \mathcal{F}_{E_+}, \mu_\beta)$,*

$$\mathbb{E}_{\mu_\beta}[F \,|\, \mathcal{F}_{E_-}] = \mathbb{E}_{\mu_\beta}[F \,|\, \mathcal{F}_{E_0}] \qquad \mu_\beta\text{-a.s.}$$

Proof. Write $\mathbb{E}$ for $\mathbb{E}_{\mu_\beta}$ and $\alpha := \alpha_*(\beta)$. First let F be bounded, measurable, and dependent only on a finite $A \subseteq E_+ \cap E_r$. For $n \geq r$ let $\Delta_n := (E_n \cap E_+) \setminus E_0$. We record three facts.

(1) *Every plaquette touching Δ_n has all its corners in $\{x_0 \geq 0\}$.* A link of Δ_n is either spatial at a time $t \geq 1$, and then the plaquettes containing it have corners at times in $\{t-1, t, t+1\} \subseteq \{x_0 \geq 0\}$, or temporal from time $s \geq 0$ to $s+1$, and then the plaquettes containing it have corners at times $s, s+1$. Hence $\partial\Delta_n \subseteq E_+$.

(2) $\partial\Delta_n \setminus E_0 \subseteq E \setminus E_n$, because $(E_+ \setminus E_0) \cap E_n = \Delta_n$ is disjoint from $\partial\Delta_n$; and $\Delta_n \cap E_- = \emptyset$, because $E_+ \cap E_- = E_0$.

(3) $A \setminus \Delta_n \subseteq E_0$, because $A \subseteq E_+ \cap E_n$.

Let $h_n := \gamma_{\Delta_n}(F|\cdot)$, a version of $\mathbb{E}[F|\mathcal{F}_{E\setminus\Delta_n}]$ (Lemma 2.9). Fix $\xi \in \Omega$ and let $\tilde{h}_n(\eta) := h_n(\eta^\xi)$, where η^ξ equals η on E_0 and ξ off E_0; $\tilde{h}_n$ is $\mathcal{F}_{E_0}$-measurable. For every η, the configurations η and η^ξ agree on $E_0 \supseteq A \setminus \Delta_n$ (by (3)), and on $\partial\Delta_n$ they can differ only on $\partial\Delta_n \setminus E_0 \subseteq E \setminus E_n$ (by (2)). By Lemma 3.6 and Lemma 2.2(c),

$$\|h_n - \tilde{h}_n\|_\infty \leq \epsilon_n := \frac{\|\delta(F)\|_1}{1-\alpha}\,\alpha^{\,n+1-r}.$$

Since $\mathcal{F}_{E_-}$ and $\mathcal{F}_{E_0}$ are contained in $\mathcal{F}_{E\setminus\Delta_n}$ (by (2)), the tower property gives $\mathbb{E}[F|\mathcal{F}_{E_-}] = \mathbb{E}[h_n|\mathcal{F}_{E_-}]$ and $\mathbb{E}[F|\mathcal{F}_{E_0}] = \mathbb{E}[h_n|\mathcal{F}_{E_0}]$; and as $\tilde{h}_n$ is $\mathcal{F}_{E_0}$- and hence $\mathcal{F}_{E_-}$-measurable, both conditional expectations differ from $\tilde{h}_n$ by at most ϵ_n almost surely. Hence $|\mathbb{E}[F|\mathcal{F}_{E_-}] - \mathbb{E}[F|\mathcal{F}_{E_0}]| \leq 2\epsilon_n$ a.s. for every $n \geq r$, and the two are equal. For general $F \in L^2(\mathcal{F}_{E_+})$ use Lemma 2.4(iv) and the L^2-continuity of conditional expectations. □

Corollary 5.4 (The time-slice chain)**.** *Let $0 < \beta < \beta_*(d)$ and let ν be the law of $U^{(0)}$ under μ_β, a Borel probability measure on Ω_s.*

(i) Every $U^{(t)}$, $t \in \mathbb{Z}$, has law ν.

(ii) For $A \in L^2(\nu)$ there is a unique $PA \in L^2(\nu)$ with $\mathbb{E}[A(U^{(1)})\,|\,\mathcal{F}_{\leq 0}] = (PA)(U^{(0)})$ a.s. The map P is a linear contraction of $L^2(\nu)$ with $P\mathbf{1} = \mathbf{1}$ and $PA \geq 0$ if $A \geq 0$.

(iii) P is self-adjoint on $L^2(\nu)$.

(iv) For $s \in \mathbb{Z}$, $t \in \mathbb{N}$ and $A, B \in L^2(\nu)$: $\mathbb{E}[A(U^{(s+t)})\,|\,\mathcal{F}_{\leq s}] = (P^t A)(U^{(s)})$ a.s., and $\mu_\beta(\overline{B(U^{(0)})}A(U^{(t)})) = \langle B, P^t A\rangle_{L^2(\nu)}$.

In words: $(U^{(t)})_{t\in\mathbb{Z}}$ is a stationary, reversible Markov chain on Ω_s with stationary law ν and transition operator P. We call ν the vacuum measure *and P the* ground-state chain *of the infinite-volume theory.*

Proof. For a lattice symmetry s' with $s'\cdot$ preserving μ_β (Theorem 4.6) and a σ-algebra $\mathcal{G}$, one has $\mathbb{E}[H \circ s'\,|\,(s')^{-1}\mathcal{G}] = \mathbb{E}[H|\mathcal{G}] \circ s'$ (both sides are $(s')^{-1}\mathcal{G}$-measurable and have the same integrals over the sets $(s')^{-1}S$, $S \in \mathcal{G}$, by invariance); here $(H \circ s')(U) := H(s' \cdot U)$. (i) The translation τ_{te_0} maps $U^{(0)}$ to $U^{(t)}$: $(\tau_{te_0} \cdot U)^{(0)} = U^{(t)}$. (ii) $A(U^{(1)}) \in L^2(\mathcal{F}_{E_+})$, so by Theorem 5.3 $\mathbb{E}[A(U^{(1)})|\mathcal{F}_{\leq 0}] = \mathbb{E}[A(U^{(1)})|\sigma(U^{(0)})]$, which is of the form $h(U^{(0)})$ for a measurable h, unique ν-a.e. (Doob–Dynkin). Put $PA := h$. Then $\|PA\|_{L^2(\nu)} \leq \|A(U^{(1)})\|_{L^2(\mu_\beta)} = \|A\|_{L^2(\nu)}$ by (i), and linearity, $P\mathbf{1} = \mathbf{1}$ and positivity are those of conditional expectations. (iii) For $A, B \in L^2(\nu)$, $\langle B, PA\rangle_\nu = \mu_\beta(\overline{B(U^{(0)})}A(U^{(1)}))$. Since $(\theta \cdot U)^{(t)} = U^{(-t)}$, θ-invariance gives $\mu_\beta(\overline{B(U^{(0)})}A(U^{(1)})) = \mu_\beta(\overline{B(U^{(0)})}A(U^{(-1)}))$, and τ_{e_0}-invariance turns this into $\mu_\beta(\overline{B(U^{(1)})}A(U^{(0)})) = \overline{\langle A, PB\rangle_\nu} = \langle PB, A\rangle_\nu$. (iv) Let $T := \tau_{se_0}$. Then $T^{-1}\mathcal{F}_{E_-} = \mathcal{F}_{\leq s}$, $T^{-1}\sigma(U^{(0)}) = \sigma(U^{(s)})$, and $A(U^{(s+1)}) = A(U^{(1)}) \circ T$. By the first paragraph and (ii), $\mathbb{E}[A(U^{(s+1)})|\mathcal{F}_{\leq s}] = (PA)(U^{(s)})$. By induction on t and the tower property ($\mathcal{F}_{\leq s} \subseteq \mathcal{F}_{\leq s+t-1}$), $\mathbb{E}[A(U^{(s+t)})|\mathcal{F}_{\leq s}] = \mathbb{E}[(PA)(U^{(s+t-1)})|\mathcal{F}_{\leq s}] = (P^{t-1}PA)(U^{(s)})$. The last formula follows by taking $s = 0$, multiplying by $\overline{B(U^{(0)})}$ and integrating. □

Remark 5.5 (Relation to the finite-volume ground-state chain). On a periodic lattice, letting $L_0 \to \infty$ at fixed spatial lattice, the time-zero marginal tends to $\Omega^2 dU$ and the transition kernel of consecutive slices is the kernel $P(U, U')$ of [F, Theorem 'Exact reformulation of the gap' (`thm:chain`)] (by [F, Theorem `thm:gs`]). Corollary 5.4 is the infinite-volume statement: the slice process of the unique Gibbs measure is itself a reversible Markov chain, and its transition operator plays the role of the ground-state transformed transfer matrix. That P is a Markov operator (and not merely a bounded operator) is a consequence of the global Markov property, which we have only at strong coupling.

6 Reflection positivity and the transfer operator

Definition 6.1 (Reflection, shifts, half-space algebras)**.** For $F : \Omega \to \mathbb{C}$ put $(\Theta F)(U) := \overline{F(\theta \cdot U)}$ and, for $t \in \mathbb{Z}$, $(S_t F)(U) := F(\tau_{te_0} \cdot U)$, so that $S_t F$ evaluates F on the configuration translated by t units of time. Let $\mathcal{A}_\pm$ be the set of bounded measurable functions depending only on finitely many links of $E_\pm$.

Lemma 6.2. *Θ is antilinear, multiplicative and involutive, and maps $\mathcal{A}_+$ onto $\mathcal{A}_-$; $S_s S_t = S_{s+t}$; $S_t \mathcal{A}_+ \subseteq \mathcal{A}_+$ for $t \geq 0$; $\Theta S_t = S_{-t}\Theta$; and $\|\delta(\Theta F)\|_1 = \|\delta(F)\|_1$.*

Proof. The action of lattice symmetries is a group action, $(ss') \cdot U = s \cdot (s' \cdot U)$ (from $(s \cdot U)(y, z) = U(s^{-1}y, s^{-1}z)$), and $\theta^2 = \mathrm{id}$, so $\Theta^2 = \mathrm{id}$; antilinearity and multiplicativity are clear. The map θ sends links with both endpoints in $\{x_0 \geq 0\}$ to links with both endpoints in $\{x_0 \leq 0\}$; τ_{te_0} with $t \geq 0$ moves the dependence of a function to later times, which preserves $\mathcal{A}_+$. As maps of $\mathbb{Z}^d$, $\theta \circ \tau_{te_0} = \tau_{-te_0} \circ \theta$, whence $\Theta S_t F(U) = \overline{F(\tau_{te_0} \cdot \theta \cdot U)} = \overline{F(\theta \cdot \tau_{-te_0} \cdot U)} = S_{-t}\Theta F(U)$. Finally $\theta\cdot$ permutes the link variables and inverts some of them, so $\delta_\ell(\Theta F) = \delta_{\theta\ell}(F)$. □

Definition 6.3 (OS measures)**.** A Borel probability measure μ on Ω is an *OS measure* if it is invariant under τ_{e_0} and under θ, and $\mu(\Theta F \cdot F) \geq 0$ for every $F \in \mathcal{A}_+$ (reflection positivity).

Proposition 6.4 (Reflection positivity in infinite volume)**.** *(a) Let $\beta \geq 0$. Every infinite-volume limit point along periodic lattices with even $L_0^{(k)}$ is an OS measure.*

(b) Let $0 < \beta < \beta_*(d)$. Then μ_β is an OS measure, and $\mu_\beta(\overline{A(U^{(0)})}\, A(U^{(1)})) \geq 0$ for every bounded measurable A on Ω_s depending on finitely many links.

Proof. (a) Invariance is Proposition 4.3(b). Let $F \in \mathcal{A}_+ \cap C_{\mathrm{loc}}$ depend only on links of $E_+ \cap E_n$. Let Λ be one of the periodic lattices of the sequence, with $L_0 = 2T$ and all sides $\geq 2n + 3$. By the proof of Proposition 4.3(b), $\theta \cdot \iota_\Lambda U' = \iota_\Lambda(\theta_\Lambda^* U')$, hence $(\Theta F) \circ \iota_\Lambda = \Theta_\Lambda(F \circ \iota_\Lambda)$, where Θ_Λ is the reflection of [F, Section 'Reflection positivity']. The function $F \circ \iota_\Lambda$ depends only on the links $\pi\ell$, $\ell \in E_+ \cap E_n$, which have both endpoints at times in $[0, n] \subseteq [0, T]$; so it belongs to the algebra $\mathcal{A}_+$ of [F], and [F, Theorem `thm:rp`] gives $\tilde{\mu}_{\Lambda,\beta}(\Theta F \cdot F) = \langle \Theta_\Lambda(F \circ \iota_\Lambda)\,(F \circ \iota_\Lambda)\rangle_{\Lambda,\beta} \geq 0$. As $\Theta F \cdot F$ is continuous, $\mu(\Theta F \cdot F) \geq 0$ for the limit point μ. For a general $F \in \mathcal{A}_+$, depending only on a finite $A \subseteq E_+$, let m_A be the law of U_A under μ; continuous functions are dense in $L^2(G^A, m_A)$, and composing them with the radial retraction of $\mathbb{C}$ onto the closed disc of radius $\|F\|_\infty$ (which is 1-Lipschitz and fixes the values of F) we obtain $F_j \in \mathcal{A}_+ \cap C_{\mathrm{loc}}$ with $\|F_j\|_\infty \leq \|F\|_\infty$ and $\|F_j - F\|_{L^2(\mu)} \to 0$. By θ-invariance $\|\Theta(F_j - F)\|_{L^2(\mu)} = \|F_j - F\|_{L^2(\mu)}$, so by the Cauchy–Schwarz inequality $\mu(\Theta F_j \cdot F_j) \to \mu(\Theta F \cdot F)$, which is therefore ≥ 0.

(b) By Theorem 4.4(iii) μ_β is the limit point of every sequence of periodic lattices, in particular of one with even $L_0^{(k)}$, so (a) applies. For the second claim let A depend only on finitely many links of E_s, and let Λ be a periodic lattice with $L_0 \geq 2$ and spatial sides so large that π is injective on the spatial links on which $A(U^{(0)})$ and $A(U^{(1)})$ depend. Let

A_Λ be the corresponding function on the spatial configurations Ω_{Λ_s} of the torus, so that $\overline{A(U^{(0)})}A(U^{(1)})\circ\iota_\Lambda = \overline{A_\Lambda}(0)\,A_\Lambda(1)$ in the notation of [F, Theorem 'Transfer matrix formalism' (`thm:tm`)]. That theorem gives

$$\tilde\mu_{\Lambda,\beta}(\overline{A(U^{(0)})}A(U^{(1)})) = \frac{\mathrm{Tr}(\hat T^{L_0-1}\hat A\,\hat T\,\hat A^*)}{\mathrm{Tr}\,\hat T^{L_0}},$$

with $\hat A$ the multiplication by A_Λ. The transfer matrix $\hat T$ is positive and trace class [F, Lemma `lem:K`](iii), so $X := \hat T^{1/2}\hat A^*\hat T^{(L_0-1)/2}$ is Hilbert–Schmidt and, by cyclicity of the trace for products of two Hilbert–Schmidt operators, $\mathrm{Tr}(\hat T^{L_0-1}\hat A\hat T\hat A^*) = \mathrm{Tr}(X^*X) \geq 0$; also $\mathrm{Tr}\,\hat T^{L_0} > 0$. By Theorem 4.4(iii) the left side converges to $\mu_\beta(\overline{A(U^{(0)})}A(U^{(1)}))$, which is therefore ≥ 0. □

Theorem 6.5 (OS reconstruction of the lattice theory)**.** *Let μ be an OS measure.*

(i) $\langle F, G\rangle := \mu(\Theta F\cdot G)$ *is a positive semidefinite Hermitian form on $\mathcal{A}_+$. Let $\mathcal{N}$ be its null space, $\mathcal{H}_\mu$ the Hilbert space completion of $\mathcal{A}_+/\mathcal{N}$, $[F]$ the class of F, and $\Omega_\mu := [1]$, a unit vector.*

(ii) For $t \in \mathbb{N}$ the map $[F] \mapsto [S_tF]$ is well defined on $\mathcal{A}_+/\mathcal{N}$ and extends to a bounded self-adjoint operator $\hat T_\mu^{(t)}$ on $\mathcal{H}_\mu$ with $\|\hat T_\mu^{(t)}\| \leq 1$; moreover $\hat T_\mu^{(t)} = (\hat T_\mu)^t$ with $\hat T_\mu := \hat T_\mu^{(1)}$, and $\hat T_\mu\Omega_\mu = \Omega_\mu$.

(iii) (Gap from clustering) *Suppose that $m > 0$ and that for every $F \in \mathcal{A}_+$ there is $C_F < \infty$ with*

$$|\mu(\Theta F\cdot S_tF) - |\mu(F)|^2| \leq C_F e^{-mt} \qquad \text{for all } t \in \mathbb{N}.$$

Then $\|\hat T_\mu|_{\Omega_\mu^\perp}\| \leq e^{-m}$; in particular ±1 are not eigenvalues of $\hat T_\mu$ on $\Omega_\mu^\perp$.

Definition 6.6 (Mass gap of an infinite-volume lattice theory)**.** For an OS measure μ, $m(\mu) := -\log\|\hat T_\mu(1 - |\Omega_\mu\rangle\langle\Omega_\mu|)\| \in [0,\infty]$, with $-\log 0 := \infty$. By Theorem 6.5(ii), $\hat T_\mu$ leaves $\Omega_\mu^\perp$ invariant, so $e^{-m(\mu)}$ is the spectral radius of $\hat T_\mu$ on $\Omega_\mu^\perp$; $m(\mu) > 0$ means that the vacuum is the only time-translation-invariant state and that the other states are separated from it by a gap.

Proof of Theorem 6.5. (i) Sesquilinearity: Θ is antilinear. Hermiticity: by θ-invariance and $\theta^2 = \mathrm{id}$, $\overline{\langle F, G\rangle} = \mu(F\circ\theta\cdot\overline{G}) = \mu(F\cdot\overline{G}\circ\theta) = \mu(\Theta G\cdot F) = \langle G, F\rangle$, where $(H\circ\theta)(U) := H(\theta\cdot U)$. Positivity is reflection positivity. $\|\Omega_\mu\|^2 = \mu(1) = 1$. (ii) For $F, G \in \mathcal{A}_+$ and $t \geq 0$, by Lemma 6.2 and τ_{e_0}-invariance, $\langle S_tF, G\rangle = \mu(S_{-t}\Theta F\cdot G) = \mu(\Theta F\cdot S_tG) = \langle F, S_tG\rangle$ (*). By the Cauchy–Schwarz inequality for the form and (*), $\|S_tF\|^2 = \langle F, S_{2t}F\rangle \leq \|F\|\,\|S_{2t}F\|$; so $F \in \mathcal{N}$ implies $S_tF \in \mathcal{N}$, and $[F] \mapsto [S_tF]$ is well defined. Iterating the same inequality with $t = 1, 2, 4, \ldots$ gives $\|S_1F\| \leq \|F\|^{1-2^{-k}}\|S_{2^k}F\|^{2^{-k}}$, and $\|S_{2^k}F\|^2 = \mu(\Theta S_{2^k}F\cdot S_{2^k}F) \leq \|F\|^2_\infty$; letting $k\to\infty$, $\|S_1F\| \leq \|F\|$ (if $\|F\| = 0$ then $\|S_1F\| = 0$ directly). So $[F] \mapsto [S_1F]$ extends to a contraction $\hat T_\mu$, self-adjoint by (*); and $S_t = S_1^t$ gives $\hat T_\mu^{(t)} = \hat T_\mu^t$. Finally $S_1 1 = 1$. (iii) Let $F \in \mathcal{A}_+$, $c := \langle\Omega_\mu, [F]\rangle = \mu(F)$ and $\psi := [F] - c\,\Omega_\mu \in \Omega_\mu^\perp$. As $\hat T_\mu$ is self-adjoint and fixes Ω_μ, $\langle\psi, \hat T_\mu^t\psi\rangle = \langle[F], \hat T_\mu^t[F]\rangle - |c|^2 = \mu(\Theta F\cdot S_tF) - |\mu(F)|^2$. Let σ_ψ be the spectral measure of $\hat T_\mu$ for ψ, supported in $[-1,1]$. For $0 < m' < m$ and even t, $\sigma_\psi(\{|\lambda| > e^{-m'}\})e^{-m't} \leq \int\lambda^t\,d\sigma_\psi \leq C_Fe^{-mt}$, so $\sigma_\psi(\{|\lambda| > e^{-m'}\}) = 0$, and hence $\sigma_\psi(\{|\lambda| > e^{-m}\}) = 0$. The vectors ψ of this form are dense in $\Omega_\mu^\perp$ (the orthogonal projection onto $\Omega_\mu^\perp$ is continuous and $\mathcal{A}_+/\mathcal{N}$ is dense in $\mathcal{H}_\mu$). The spectral projection $E(\{|\lambda| > e^{-m}\})$ of $\hat T_\mu$ commutes with the projection onto $\Omega_\mu^\perp$ and vanishes on a dense subset of $\Omega_\mu^\perp$, hence on all of it, and $\|\hat T_\mu|_{\Omega_\mu^\perp}\| \leq e^{-m}$. □

Remark 6.7 (The mass gap conjecture in infinite volume). By Theorem 1.2, the following statement is meaningful at every coupling: *for $G = SU(N)$, $d = 4$ and every $\beta > 0$, every infinite-volume limit point μ (along periodic lattices with even L_0) has $m(\mu) > 0$.* This is the infinite-volume form of conjecture (G1) of the research program. Its relation to the finite-volume formulation $\inf_{\Lambda_s} m_{\Lambda_s}(\beta) > 0$ of [F] is not addressed here; the two differ in the order in which the time extent and the spatial extent are sent to infinity. At strong coupling both hold (Theorem 7.1 and Corollary 3.4). For $G = U(1)$, $d = 4$ and large β the analogous statement is false [Gut80, FS82].

7 The mass gap of the infinite-volume theory at strong coupling

Theorem 7.1 (Transfer operator, ground-state chain and mass gap)**.** *Let $d \geq 2$, $0 < \beta < \beta_*(d)$, $\alpha := \alpha_*(\beta)$, $\mu := \mu_\beta$, and let ν and P be as in Corollary 5.4.*

(i) For $F \in \mathcal{A}_+$ let $\hat{F} \in L^2(\nu)$ be defined by $\hat{F}(U^{(0)}) = \mathbb{E}_\mu[F|\mathcal{F}_{E_0}]$. Then $[F] \mapsto \hat{F}$ is well defined on $\mathcal{A}_+/\mathcal{N}$ and extends to a unitary operator $W : \mathcal{H}_\mu \to L^2(\nu)$ with $W\Omega_\mu = \mathbf{1}$ and $W\hat{T}_\mu W^{-1} = P$.

(ii) $0 \leq P \leq 1$; hence $\hat{T}_\mu \geq 0$.

(iii) $\operatorname{spec} P \subseteq [0, \alpha] \cup \{1\}$, and 1 is a simple eigenvalue with eigenvector $\mathbf{1}$. Equivalently, $\operatorname{gap} P := 1 - \sup \operatorname{spec}(P|_{\mathbf{1}^\perp}) \geq 1 - \alpha$ (with $\sup \emptyset := 0$), and

$$m(\mu_\beta) = -\log(1 - \operatorname{gap} P) \;\geq\; \log \frac{1}{\alpha_*(\beta)}.$$

(iv) For all $A, B \in L^2(\nu)$ and $t \in \mathbb{N}$, $|\mu(\overline{B(U^{(0)})}A(U^{(t)})) - \overline{\nu(B)}\nu(A)| \leq \alpha^t \|B - \nu(B)\|_{L^2(\nu)} \|A - \nu(A)\|_{L^2(\nu)}$.

Proof. Write $\mathbb{E} := \mathbb{E}_\mu$. We use the transport identity of the proof of Corollary 5.4: for a measure-preserving lattice symmetry T, $\mathbb{E}[H \circ T|T^{-1}\mathcal{G}] = \mathbb{E}[H|\mathcal{G}] \circ T$. With $T = \tau_{se_0}$ and Theorem 5.3 it gives the Markov property at time s: $\mathbb{E}[H|\mathcal{F}_{\leq s}] = \mathbb{E}[H|\sigma(U^{(s)})]$ for $H \in L^2(\mathcal{F}_{\geq s})$.

(i) *Reflection.* For $F \in \mathcal{A}_+$, $\mathbb{E}[\Theta F|\mathcal{F}_{E_0}] = \overline{\hat{F}(U^{(0)})}$: for bounded $\mathcal{F}_{E_0}$-measurable H, θ-invariance gives $\mu(H \cdot \Theta F) = \mu((H \circ \theta) \cdot \overline{F})$, and $H \circ \theta = H$ because θ fixes every link of E_0; so $\mu(H \cdot \Theta F) = \mu(H\overline{F}) = \mu(H\overline{\hat{F}(U^{(0)})})$. *Isometry.* For $F, G \in \mathcal{A}_+$, ΘF is $\mathcal{F}_{E_-}$-measurable, so by Theorem 5.3

$$\langle F, G\rangle = \mu(\Theta F \cdot \mathbb{E}[G|\mathcal{F}_{E_-}]) = \mu(\Theta F \cdot \hat{G}(U^{(0)})) = \mu(\mathbb{E}[\Theta F|\mathcal{F}_{E_0}]\, \hat{G}(U^{(0)})) = \langle \hat{F}, \hat{G}\rangle_{L^2(\nu)}.$$

Hence $[F] \mapsto \hat{F}$ is well defined and isometric, and extends to an isometry W of $\mathcal{H}_\mu$ into $L^2(\nu)$. Its range is closed and contains every bounded measurable function A of finitely many links of E_s (for which $A(U^{(0)}) \in \mathcal{A}_+$ and $\hat{A} = A$); these are dense in $L^2(\nu)$, by Lemma 2.4(iv) (for the link set E_0) and the unitary map $L^2(\nu) \to L^2(\Omega, \mathcal{F}_{E_0}, \mu)$, $B \mapsto B(U^{(0)})$. So W is unitary, and $W\Omega_\mu = \hat{1} = \mathbf{1}$. *Intertwining.* Let $F \in \mathcal{A}_+$. The function $S_1 F = F \circ \tau_{e_0}$ is $\mathcal{F}_{\geq 1}$-measurable, so by the Markov property at time 1 and the transport identity, $\mathbb{E}[S_1 F|\mathcal{F}_{\leq 1}] = \mathbb{E}[F \circ \tau_{e_0}|\sigma(U^{(1)})] = \hat{F}(U^{(1)})$. By the tower property ($\mathcal{F}_{E_0} \subseteq \mathcal{F}_{\leq 0} \subseteq \mathcal{F}_{\leq 1}$) and Corollary 5.4(ii),

$$\mathbb{E}[S_1 F|\mathcal{F}_{E_0}] = \mathbb{E}\big[\mathbb{E}[\hat{F}(U^{(1)})|\mathcal{F}_{\leq 0}]\,\big|\,\mathcal{F}_{E_0}\big] = (P\hat{F})(U^{(0)}),$$

i.e. $W\hat{T}_\mu[F] = PW[F]$. Both sides are bounded operators, and $\mathcal{A}_+/\mathcal{N}$ is dense, so $W\hat{T}_\mu = PW$.

(ii) P is a self-adjoint contraction (Corollary 5.4). For bounded measurable A depending on finitely many links of E_s, $\langle A, PA\rangle_\nu = \mu(\overline{A(U^{(0)})}A(U^{(1)})) \geq 0$ by Proposition 6.4(b); such A are dense in $L^2(\nu)$ and $A \mapsto \langle A, PA\rangle_\nu$ is continuous, so $P \geq 0$. By (i), $\hat{T}_\mu \geq 0$.

(iii) Let $F \in \mathcal{A}_+$ depend only on a finite $A \subseteq E_+$. By θ- and τ_{e_0}-invariance, $\mu(\Theta F) = \overline{\mu(F)}$ and $\mu(S_t F) = \mu(F)$, so $\mu(\Theta F \cdot S_t F) - |\mu(F)|^2 = \mathrm{Cov}_\mu(\Theta F, S_t F)$. The function ΘF depends only on the links $\theta\ell$, $\ell \in A$, all of whose endpoints have time coordinate ≤ 0; $S_t F$ depends only on the links $\ell + te_0$, $\ell \in A$, all of whose endpoints have time coordinate $\geq t$. For $t \geq 1$ these sets are disjoint and at distance $\geq t$ (Lemma 2.2(b)), so Theorem 5.1 and Lemma 6.2 give $|\mathrm{Cov}_\mu(\Theta F, S_t F)| \leq \|F\|_\infty \|\delta(F)\|_1 \alpha^t/(1-\alpha)$; for $t = 0$ the left side is at most $2\|F\|_\infty^2$. Theorem 6.5(iii) with $m = \log(1/\alpha)$ gives $\|\hat{T}_\mu|_{\Omega_\mu^\perp}\| \leq \alpha$, hence by (i) $\|P|_{\mathbf{1}^\perp}\| \leq \alpha$, and by (ii) $\mathrm{spec}(P|_{\mathbf{1}^\perp}) \subseteq [0, \alpha]$. Since $P\mathbf{1} = \mathbf{1}$ and P is self-adjoint, $\mathbb{C}\mathbf{1}$ and $\mathbf{1}^\perp$ reduce P; so $\mathrm{spec}\, P \subseteq [0, \alpha] \cup \{1\}$ and, as $\alpha < 1$, the eigenspace of 1 is $\mathbb{C}\mathbf{1}$. For a positive operator the norm is the supremum of the spectrum, so $e^{-m(\mu)} = \|\hat{T}_\mu|_{\Omega_\mu^\perp}\| = \|P|_{\mathbf{1}^\perp}\| = 1 - \mathrm{gap}\, P$.

(iv) Write $A = \nu(A)\mathbf{1} + A_0$, $B = \nu(B)\mathbf{1} + B_0$ with $A_0, B_0 \perp \mathbf{1}$. Then $P^t A = \nu(A)\mathbf{1} + P^t A_0$ with $P^t A_0 \perp \mathbf{1}$, and by Corollary 5.4(iv) $\mu(\overline{B(U^{(0)})}A(U^{(t)})) = \langle B, P^t A\rangle_\nu = \overline{\nu(B)}\nu(A) + \langle B_0, P^t A_0\rangle_\nu$, with $|\langle B_0, P^t A_0\rangle_\nu| \leq \alpha^t \|B_0\|\|A_0\|$ by (iii). □

Proposition 7.2 (Gauge invariance of the ground-state chain)**.** *Let $0 < \beta < \beta_*(d)$. For $g \in G^{\mathbb{Z}^{d-1}}$ let g act on Ω_s by $(g \cdot u)_{(\vec{x},i)} = g(\vec{x}) u_{(\vec{x},i)} g(\vec{x} + e_i)^{-1}$, and let $\mathcal{H}_{\mathrm{phys}} \subseteq L^2(\nu)$ be the closed subspace of functions A with $A(g \cdot u) = A(u)$ for ν-a.e. u, for every g. Then ν is invariant under every such g, P maps $L^2(\nu)$ into $\mathcal{H}_{\mathrm{phys}}$, and P vanishes on $\mathcal{H}_{\mathrm{phys}}^\perp$. In particular the spectral statements of Theorem 7.1 concern the physical (gauge-invariant) sector, and gauge-variant states lie in* $\ker P$*, as in finite volume (where $\hat{T} = P\hat{T}P$, [F, Lemma `lem:K`](iii)).*

Proof. For $s \in \mathbb{Z}$ let $g^{[s]} \in G^{\mathbb{Z}^d}$ equal g on the slice $x_0 = s$ and 1 elsewhere. Then $(g^{[s]} \cdot U)^{(s)} = g \cdot U^{(s)}$ and $(g^{[s]} \cdot U)^{(t)} = U^{(t)}$ for $t \neq s$ (only temporal links touching the slice, and its spatial links, are changed). By Theorem 4.6, μ_β is $g^{[s]}$-invariant; with $s = 0$ this gives the invariance of ν. For $A, B \in L^2(\nu)$, $g^{[1]}$-invariance gives $\langle B, P(A \circ g)\rangle_\nu = \mu_\beta(\overline{B(U^{(0)})}A(g \cdot U^{(1)})) = \langle B, PA\rangle_\nu$, so $P(A \circ g) = PA$, where $(A \circ g)(u) := A(g \cdot u)$; and $g^{[0]}$-invariance gives $\langle B, PA\rangle_\nu = \langle B \circ g, PA\rangle_\nu = \langle B, (PA) \circ g^{-1}\rangle_\nu$ (using the invariance of ν), so $PA = (PA) \circ g^{-1}$. Thus $PA \in \mathcal{H}_{\mathrm{phys}}$. If $A \perp \mathcal{H}_{\mathrm{phys}}$ then $\langle B, PA\rangle_\nu = \langle PB, A\rangle_\nu = 0$ for all B, i.e. $PA = 0$. □

Remark 7.3 (The lattice OS axioms). For $0 < \beta < \beta_*(d)$ the unique Gibbs measure μ_β satisfies the lattice versions of the Osterwalder–Schrader axioms for all bounded local observables: invariance under lattice translations and the hypercubic group (Theorem 4.6); reflection positivity for the time reflection through a hyperplane of sites (Proposition 6.4), hence by hypercubic invariance for the reflection through any coordinate hyperplane of sites; and exponential clustering (Theorem 5.1). The reconstructed theory (Theorem 6.5) has a positive transfer operator, a unique vacuum, and a mass gap (Theorem 7.1). This is milestone T1.1 of the research program, for $\beta < \beta_*(d)$ (a larger range than the $\beta < \beta_0(d)$ of the milestone's statement).

Corollary 7.4 (Area law in infinite volume)**.** *Let $0 < \beta < \beta_0(d)$, the radius of the cluster expansion of [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)]. Then $\beta < \beta_*(d)$, and for every irreducible representation r with non-zero z_0-charge (e.g. $G = SU(N)$ and N-ality $n(r) \neq 0$) and every rectangular loop C in $\mathbb{Z}^d$ with sides $R, T \geq 1$,*

$$\left|\mu_\beta(\chi_r(U_C))\right| \leq d_r \exp(-\sigma(\beta)RT + \kappa(\beta)|C|),$$

with $\sigma(\beta), \kappa(\beta)$ as in [III, Definition 'Constants' (`al:def:constants`)].

Proof. $\beta_0(d) \leq \frac{1}{2e^2K(K+2)} < \frac{1}{2e^2K^2} < \frac{1}{6(d-1)} \leq \mathrm{artanh}\, \frac{1}{6(d-1)} = \beta_*(d)$, with $K = 4(2d-3) \geq 4$, since $\log(1+y) \leq y$, $2e^2K^2 \geq 2e^2 \cdot 16(d-1)^2 > 6(d-1)$ and $\mathrm{artanh}\, y \geq y$. By Theorem 4.4(iii),

μ_β is an infinite-volume limit point, and [III, Corollary 'String tension' (`al:cor:tension`)](ii) applies to it. □

8 Discussion

Remark 8.1 (The $U(1)$ test). The proofs use only: compactness and metrisability of G; the bound $|\log w_\beta| \le \beta$ and the facts that w_β is a continuous class function invariant under inversion; and, for the positivity of P, the positivity of the finite-volume transfer matrix [F, Lemma `lem:K`], which rests on $c_r(\beta) \ge 0$. All of these hold for $G = U(1)$ with $\rho = \mathrm{id}$. Hence every statement of this article holds verbatim for $U(1)$ (and $U(N)$, and finite groups) for $\beta < \beta_*(d)$. For $U(1)$ in $d = 4$ the mass gap is known to vanish at large β (massless photon) [Gut80, FS82]; so, exactly as for [F, Theorem `thm:strong`] and for the area law of [III], no refinement of these arguments can reach the large-β regime, and the infinite-volume theory constructed here is the strong-coupling endpoint of the problem, not a step across the crossover (`docs/03-research-program.md`, §3.4). What the article does provide for the large-β problem is its formulation: by Theorem 1.2 and Definition 6.6 the infinite-volume mass gap $m(\mu)$ is defined at every coupling (Remark 6.7), and at strong coupling it is exactly the spectral gap of a reversible Markov chain of time slices of the vacuum (Theorem 7.1), in line with §3.5 of the research program.

Remark 8.2 (The thresholds). $\beta_*(4) = \operatorname{artanh} \frac{1}{18} \approx 0.0556$ in the normalisation $w_\beta = \exp(\frac{\beta}{N} \mathrm{Re}\,\mathrm{Tr}\, U)$ of [F], independent of N. For $SU(2)$ the strong-coupling series describes the Monte Carlo data up to $\beta \approx 1.5$–2 and the crossover is at $\beta \approx 2.2$ (`results/`; `docs/02-state-of-the-art.md`); for $SU(3)$ at $\beta \approx 5.6$. Any argument of Dobrushin type has a threshold of order $1/(6(d-1))$, because it compares single-link conditional distributions without using cancellations. Going further requires more structure: block criteria of Dobrushin–Shlosman type, or the Bakry–Émery approach of [SZZ23], which uses the positive Ricci curvature of $SU(N)$ (and so does not apply to $U(1)$ in the same form). We have not compared those thresholds with $\beta_*(d)$. (Heuristic:) no criterion that is checked one coupling scale at a time, without a renormalisation-group step, is expected to reach the crossover.

Remark 8.3 (Consequences for the numerics). For $A \in L^2(\nu) \ominus \mathbb{C}\mathbf{1}$ let $c_t := \langle A, P^t A\rangle_\nu = \int_{[0,1]} \lambda^t \, d\sigma_A(\lambda)$, with σ_A the spectral measure of P. Since $P \ge 0$, $c_t^2 \le c_{t-1} c_{t+1}$ by the Cauchy–Schwarz inequality, so the effective masses $m_{\mathrm{eff}}(t) := \log(c_t/c_{t+1})$ (when $c_{t+1} > 0$) are non-increasing in t and converge to $-\log \sup \operatorname{supp} \sigma_A \ge m(\mu_\beta)$. Every effective mass is therefore an *upper* bound for the mass gap. This is the infinite-volume, strong-coupling counterpart of the statement used in `lattice/scripts/analyze_gevp.py` for the finite-volume transfer matrix; at the couplings of the glueball runs ($\beta \ge 2.3$ for $SU(2)$) the finite-volume version is what applies.

Remark 8.4 (Next steps). (i) T1.2: the Hamiltonian (continuous-time) version, where the gap becomes the Poincaré constant of the Dirichlet form $\frac{g^2}{2} \sum_\ell \int |\nabla_\ell \varphi|^2 d\nu$. (ii) Whether $m(\mu)$ for limit points and $\inf_{\Lambda_s} m_{\Lambda_s}$ can be compared in general (Remark 6.7). (iii) The global Markov property was derived from uniqueness with a rate; whether it holds for all limit points at every β is open here, and it is what makes P a Markov operator.

Summary of results

Statement	Where	Status
Total variation $\le \tanh(\delta/2)$ for density ratios in $[ce^{-\delta}, ce^{\delta}]$ (sharp)	Lem. 3.1	Theorem
Dobrushin constant $\alpha_*(\beta) = 6(d-1)\tanh\beta$; finite-volume gap $\ge \log(1/\alpha_*)$	Lem. 3.3, Cor. 3.4	Theorem
Boundary influence with disagreement sets	Lem. 3.5, 3.6	Theorem
Limit points are Gibbs, translation and reflection invariant, RP (all β)	Prop. 4.3, 6.4(a)	Theorem
OS reconstruction; mass gap $m(\mu)$ defined (all β)	Thm. 6.5, Def. 6.6	Theorem
Uniqueness; convergence of periodic/free/fixed b.c. with rate ($\beta < \beta_*$)	Thm. 4.4	Theorem
Invariance under lattice symmetries and all gauge transformations; forests	Thm. 4.6, Prop. 4.7	Theorem
Exponential clustering in infinite volume	Thm. 5.1	Theorem
Global Markov property; time slices form a reversible Markov chain	Thm. 5.3, Cor. 5.4	Theorem
$\mathcal{H} \cong L^2(\nu)$, $\hat{T} \cong P \ge 0$; $\operatorname{spec} P \subseteq [0,\alpha_*] \cup \{1\}$; $m(\mu_\beta) \ge \log(1/\alpha_*)$	Thm. 7.1	Theorem
Gauge-variant states lie in $\ker P$	Prop. 7.2	Theorem
Area law for μ_β ($\beta < \beta_0$), using [III]	Cor. 7.4	Theorem
(G1) in infinite volume: $m(\mu) > 0$ for all limit points and all β	Rem. 6.7	open

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